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Biology
Class 11

Sindh Textbook Board

(English Medium)

Based on the Sindh Textbook Board Book

Unit 1
Biological Molecules

Unit 2
Enzymes

Unit 3
Cell Structure and Function

Unit 4
Bioenergetics

Unit 5
Acellular Life

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Prokaryotes

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Protoctists and Fungi

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Biological Molecules

UNIT 1

Introduction to Biological Molecules and Bio-elements

LESSON 1

1. Biological molecules and bio-elements

Six bio-elements make about 98% of the living system. These elements form carbohydrates, proteins, lipids and nucleic acids.

- **Carbon:** one of the six bio-elements that make about 98% of the living system.
- **Hydrogen:** one of the six bio-elements that make about 98% of the living system.
- **Oxygen:** one of the six bio-elements that make about 98% of the living system.
- **Nitrogen:** one of the six bio-elements that make about 98% of the living system.
- **Phosphorus:** one of the six bio-elements that make about 98% of the living system.
- **Sulphur:** one of the six bio-elements that make about 98% of the living system.

2. Other essential elements

The remaining elements are present in small amounts but are also essential.

- **Calcium**
- **Potassium**
- **Sodium**
- **Chlorine**
- **Magnesium**
- **Iron**

Group	Share / presence
Carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur	About 98% of the living system
Calcium, potassium, sodium, chlorine, magnesium and iron	Present in small amounts, also essential

3. Polymers and hydrolysis

A polymer of 21 monomers has 20 bonds between them, so 20 water molecules are needed to completely hydrolyse it and break all the bonds.

Chain: 21 monomers → 20 bonds between them → 20 water molecules needed for complete hydrolysis.

Water — Importance and Properties

LESSON 2

1. Hydrogen bond between water molecules

The weak attraction between the partial positive hydrogen of one water molecule and the partial negative oxygen of another is a hydrogen bond. The slight negative charge at one end of one water molecule is attracted to the slight positive charge of another water molecule.

2. Hydrophobic exclusion

Non-polar oil molecules are excluded by water and come together, which is called hydrophobic exclusion. This is the tendency of water to coalesce oil drop into large droplet.

3. Why water is called amphoteric

Water is called amphoteric because it can behave both as an acid and as a base. Due to ionization, a water molecule can donate a hydrogen ion (H^+) and act as an acid, or accept a hydrogen ion and act as a base.

This property helps water maintain pH and take part in many biochemical reactions in the cell.

- **As an acid:** water donates a hydrogen ion (H^+).
- **As a base:** water accepts a hydrogen ion.
- **Role:** helps maintain pH and takes part in many biochemical reactions in the cell.

Property	Meaning
Hydrogen bond	Weak attraction between partial positive hydrogen of one water molecule and partial negative oxygen of another
Hydrophobic exclusion	Non-polar oil molecules are excluded by water and come together
Amphoteric	Water can behave both as an acid and as a base

Carbohydrates

LESSON 3

1. How monosaccharides are classified

Monosaccharides are classified on the basis of the number of carbon atoms and the type of carbonyl group.

- **Trioses:** 3C.
- **Tetroses:** 4C.
- **Pentoses:** 5C, like ribose.
- **Hexoses:** 6C, like glucose.
- **Aldoses:** having an aldehyde group.
- **Ketoses:** having a ketone group.

2. Glycosidic bond

Two monosaccharides are joined by a glycosidic bond formed by condensation. The covalent bond between two monosaccharides is called a glycosidic bond.

3. Starch and cellulose

Both starch and cellulose are polymers of glucose; they differ in the type of glycosidic linkage. Cellulose is the main structural carbohydrate of plant cell walls and is the most abundant carbohydrate in nature.

4. Properties and roles of starch, glycogen, cellulose and chitin

- **Starch:** the storage polysaccharide of plants, made of glucose units; it stores energy in seeds and tubers which can be broken down when needed.
- **Glycogen:** the storage polysaccharide of animals, highly branched, stored in liver and muscles and quickly converted to glucose.
- **Cellulose:** a structural polymer of glucose with beta linkages; it forms plant cell walls and gives strength and support, and humans cannot digest it.
- **Chitin:** a structural polysaccharide containing nitrogen, found in fungal cell walls and the exoskeleton of arthropods, where it provides protection and support.

Polysaccharide	Role
Starch	Storage in plants — seeds and tubers
Glycogen	Storage in animals — liver and muscles
Cellulose	Structure — plant cell walls
Chitin	Structure — fungal cell walls and exoskeleton of arthropods

Proteins

LESSON 4

1. Importance of proteins

Proteins are the most important organic compound of the cell which carry out virtually all of the cell's activities. Proteins work as enzymes, hormones, transporters and structural parts and carry out almost all cell activities.

2. Why amino acids are named so

Amino acids are named so because each molecule contains both an amino group (-NH2) and a carboxylic acid group (-COOH). These two groups are attached to the same central alpha carbon atom. The amino group gives basic character and the carboxyl group gives acidic character, so the name combines both groups.

3. Structure of an amino acid

An amino acid is an organic compound containing an amino group, a carboxyl group, a hydrogen atom and a variable R group attached to a central alpha carbon. About 20 different amino acids make all proteins.

Every amino acid has the same amino group, carboxyl group and alpha carbon; only the radical (R) group differs.

4. Peptide linkage formation

A peptide linkage is formed when the carboxyl group of one amino acid reacts with the amino group of another amino acid. A molecule of water is removed in this condensation reaction, and the resulting -CO-NH- bond is the peptide bond. Repeating this process joins many amino acids into a polypeptide chain.

Chain: amino acid + amino acid → peptide bond (-CO-NH-) + water removed → polypeptide chain.

5. Classification of protein

Proteins are classified on the basis of structure into fibrous and globular proteins, and on the basis of composition into simple, conjugated and derived proteins. They are also classified by function.

- **Structural proteins:** build body parts, e.g. collagen in connective tissue, keratin in hair and nails, and elastin in ligaments.
- **Functional proteins:** perform specific jobs, e.g. enzymes catalyse reactions, haemoglobin transports oxygen, antibodies give protection and hormones like insulin regulate body activities.

Lipids

LESSON 5

1. What are lipids

Lipids are a diverse group of organic compounds that are insoluble in water but soluble in organic solvents, and they include fats, oils, phospholipids and steroids.

2. Glycerol and acylglycerols

Glycerol is a three-carbon alcohol with three hydroxyl groups — a trihydroxy alcohol made of three carbon atoms — and forms the backbone of acylglycerols. Acylglycerols are formed by combining glycerol with one to three fatty acids; triglycerides store energy, provide insulation and protect organs.

3. Why fats provide more energy than carbohydrates

Fats provide more energy than carbohydrates because fat molecules contain a much higher proportion of carbon-hydrogen bonds and very little oxygen. On complete oxidation, one gram of fat gives about 9 kcal of energy, while one gram of carbohydrate gives about 4 kcal. Fats are also stored in a water-free form, so they store more energy in less space.

4. Saturated and unsaturated acylglycerols

Saturated acylglycerols	Unsaturated acylglycerols
Fatty acids with only single bonds between carbon atoms, fully loaded with hydrogen	One or more double bonds in their fatty acids, which cause bends in the chains
Usually solid at room temperature, e.g. animal fats	Usually liquid at room temperature, e.g. plant oils

5. Waxes

Waxes are esters of long-chain fatty acids with long-chain alcohols. They form waterproof coatings on leaves, fruits, skin and feathers, reducing water loss and giving protection.

6. Terpenoids

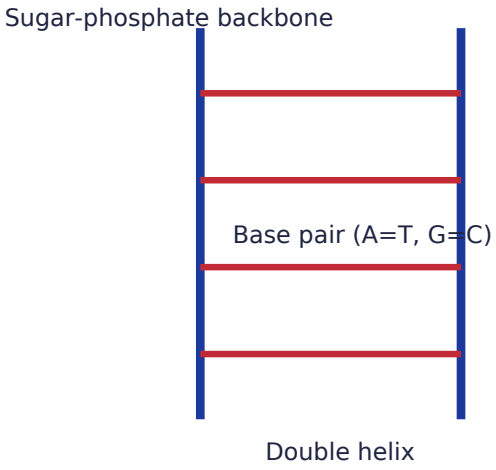
Terpenoids are a large and important class of lipids made up of repeating isoprenoid units (C_5H_8). They are classified according to the number of these units, such as monoterpenes, sesquiterpenes, diterpenes, triterpenes and tetraterpenes. Carotenoids, which are tetraterpenes, act as photosynthetic pigments and give colour to carrots and leaves. Other terpenoids include rubber, steroids precursors, essential oils and vitamins A, E and K, showing their wide biological importance.

Nucleic Acids

LESSON 6

1. What is nucleic acid

Nucleic acids are polymers of nucleotides that store and transfer genetic information; DNA and RNA are the two types.



2. How many steps involve in nucleotide formation

Nucleotide formation involves joining of three components in steps. First a nitrogenous base is attached to a pentose sugar (ribose or deoxyribose) to form a nucleoside. Then one or more phosphate groups are attached to the sugar to make a nucleotide.

Chain: base + sugar → nucleoside → + phosphate → nucleotide. Thus base + sugar + phosphate are combined step by step to complete a nucleotide.

3. Mononucleotide — ATP

ATP is a mononucleotide consisting of adenine base, ribose sugar and three phosphate groups; its high-energy phosphate bonds store and release energy for cell activities.

- **Base:** adenine.

- **Sugar:** ribose.
- **Phosphate:** three phosphate groups.

4. Dinucleotide — NAD

NAD is a dinucleotide formed by joining two nucleotides, one containing adenine and the other nicotinamide, linked through their phosphate groups. NAD works as a coenzyme and carries hydrogen and electrons during respiration.

Feature	ATP	NAD
Type	Mononucleotide	Dinucleotide
Composition / parts	Adenine base, ribose sugar and three phosphate groups	Two nucleotides — one containing adenine and the other nicotinamide, linked through their phosphate groups
Role	High-energy phosphate bonds store and release energy for cell activities	Coenzyme; carries hydrogen and electrons during respiration

Conjugated Molecules

LESSON 7

1. What are conjugated molecules

Conjugated molecules are formed when biomolecules of two different groups combine chemically and act as one unit.

2. Glycolipids

Glycolipids are conjugates of lipids and carbohydrates. They are found in brain white matter, myelin sheath and membranes.

- **Components:** lipids + carbohydrates.
- **Found in:** brain white matter, myelin sheath and membranes.

3. Glycoproteins

Glycoproteins are conjugates of carbohydrate and protein. They work as receptors, transport proteins and blood group antigens.

- **Components:** carbohydrate + protein.
- **Roles:** receptors, transport proteins and blood group antigens.

4. Lipoproteins

Lipoproteins are conjugates of lipids and proteins. They transport lipids in blood plasma.

- **Components:** lipids + proteins.
- **Role:** transport lipids in blood plasma.

5. Nucleoproteins

Nucleoproteins are conjugates of nucleic acid and basic protein. They form chromatin, chromosomes and ribosomes.

- **Components:** nucleic acid + basic protein.
- **Form:** chromatin, chromosomes and ribosomes.

Conjugated molecule	Components	Role / location
Glycolipids	Lipids + carbohydrates	Brain white matter, myelin sheath and membranes
Glycoproteins	Carbohydrate + protein	Receptors, transport proteins and blood group antigens
Lipoproteins	Lipids + proteins	Transport lipids in blood plasma
Nucleoproteins	Nucleic acid + basic protein	Form chromatin, chromosomes and ribosomes

Enzymes

UNIT 2

Enzyme Structure

LESSON 8

1. What is an Enzyme?

An enzyme is a biological catalyst, usually a protein, that speeds up a specific chemical reaction in living organisms without being used up itself. Enzymes work in small amounts and remain unchanged at the end of the reaction. They need an optimum temperature and pH for maximum activity, lower the activation energy of reactions, and their activity can be increased by activators or decreased by inhibitors.

2. Structure of an Enzyme

Most enzymes are globular proteins made of one or more polypeptide chains folded into a specific three-dimensional shape. A small region called the active site is where the substrate binds; it has a binding site and a catalytic site. The active site binds the substrate and takes part directly in the catalytic reaction, and it has a definite shape and chemical nature.

3. Simple and Conjugated Enzymes

- **Simple enzymes:** Enzymes that are simple proteins.
- **Conjugated enzymes:** Enzymes consisting of a protein part called apoenzyme and a non-protein cofactor.
- **Cofactor:** The non-protein part; it may be a metal ion, a coenzyme or a tightly bound prosthetic group.
- **Holoenzyme:** The complete active form of a conjugated enzyme.

4. Prosthetic Group

A prosthetic group is a non-protein part that is permanently and tightly attached to an enzyme and is essential for its activity. It may be an organic molecule or a metal ion. Unlike a coenzyme, it does not separate from the enzyme during the reaction. Haem group in some enzymes is a common example of a prosthetic group.

5. Why Enzymes are Specific

Enzymes are specific because each enzyme has an active site of a definite shape and chemical nature. Only a substrate with a complementary shape can fit into that active site and form an enzyme-substrate complex. Therefore one enzyme generally catalyses only one type of reaction.

Mechanism of Enzyme Action

LESSON 9

1. Mechanism of Enzyme Action

In enzyme action, the substrate first binds to the active site of the enzyme to form an enzyme-substrate complex. Binding stresses the substrate bonds and lowers activation energy, converting the substrate into products. An enzyme-product complex is formed, the products are released, and the free enzyme is ready to be used again.

- **Step 1:** Substrate binds to the active site.
- **Step 2:** Enzyme-substrate complex is formed.
- **Step 3:** Substrate is converted into products.
- **Step 4:** Products are released and the enzyme is free for reuse.

2. Lock and Key Model

The lock and key model says the active site has a fixed shape complementary to the substrate.

3. Induced Fit Model

The induced fit model says the active site changes shape slightly on binding. In induced fit, the binding of the substrate changes the shape of the enzyme's active site so that it fits the substrate more closely.

4. How Enzymes Lower Activation Energy

Enzymes work by dramatically decreasing the activation energy needed for a reaction. An enzyme reduces activation energy by binding the substrate at its active site and holding it in the correct position and stressed form. This brings reacting groups close together and weakens the bonds that are to be broken. The substrate therefore reaches the transition state easily with less energy, so the reaction proceeds faster at body temperature.

5. Effect on Free Energy and Equilibrium

Enzymes speed up a reaction by lowering activation energy, but they do not change the free energy change of the reaction. An enzyme only speeds up reaching equilibrium; it cannot change the position of equilibrium. If an enzyme is added where substrate and product are in equilibrium, nothing changes and the reaction stays at equilibrium.

Factors Affecting Enzyme Action

LESSON 10

1. Effect of Temperature

Temperature strongly affects the rate of enzyme action. At low temperature enzyme molecules have little kinetic energy, so fewer collisions occur and the rate is slow. As temperature rises, collisions increase and the rate roughly doubles with every 10°C rise until the optimum temperature is reached. For most human enzymes the optimum is about 37°C. Beyond the optimum, heat denatures the enzyme by destroying its three-dimensional shape, and the rate falls sharply to zero.

2. Why Enzymes are Temperature Sensitive

Enzymes are proteins, so their activity depends strongly on temperature. At low temperature they work slowly, and activity increases up to an optimum temperature, about 37°C for human enzymes. Above the optimum, high temperature breaks the bonds holding the enzyme's shape and denatures it, so activity falls rapidly. Hot-spring bacteria have enzymes with a high optimum temperature.

3. Effect of Substrate Concentration

At low substrate concentration, increasing the substrate increases the rate because more active sites are occupied. As concentration rises, more enzyme-substrate complexes form and product formation speeds up. Gradually the active sites become occupied and the rate rises more slowly. When all active sites are saturated, the enzyme works at maximum rate (Vmax) and the rate stays constant. Further increase in substrate then has no effect.

4. Effect of Enzyme Concentration

Enzyme activity increases with enzyme concentration because more enzyme molecules provide more active sites for the substrate. When substrate is available in excess, every increase in enzyme molecules increases the number of enzyme-substrate complexes formed per unit time. As a result, more product is formed and the rate rises in direct proportion, until the substrate becomes limiting.

Enzyme Inhibition

LESSON 11

1. Enzyme Inhibition

Enzyme inhibition is the decrease or stoppage of enzyme activity by a substance called an inhibitor.

2. Competitive Inhibition

In competitive inhibition the inhibitor resembles the substrate and competes for the active site. It can be overcome by increasing substrate concentration, because the inhibitor competes for the active site and adding more substrate can overcome the inhibition. Competitive inhibition depends on the active site acting as a specific template whose shape the inhibitor mimics to block the substrate.

3. Non-competitive Inhibition

In non-competitive inhibition the inhibitor binds at an allosteric site, away from the active site, and changes the enzyme shape, so it cannot be overcome by substrate. A noncompetitive inhibitor binds away from the active site and changes the shape of the enzyme, so the reactant can no longer be acted upon effectively.

Competitive Inhibition	Non-competitive Inhibition
Inhibitor competes with substrate for the active site	Inhibitor binds at another (allosteric) site

Competitive Inhibition	Non-competitive Inhibition
Can be overcome by increasing substrate concentration	Cannot be overcome by substrate

4. Feedback Inhibition

Feedback inhibition occurs when the end product inhibits the first enzyme of its pathway. For example, in the pathway X → Y → Z → A, product A binds to the enzyme that converts X to Y at a position remote from its active site and decreases its activity; X is the substrate of the first enzyme and A is the allosteric (feedback) inhibitor.

5. Activators and Inhibitors

Activator	Inhibitor
Increases the activity of an enzyme, often by helping the substrate bind or by removing an inhibitor	Decreases or stops enzyme activity
Speeds up reactions	Slows down reactions

6. Significance of Inhibition

Inhibition is significant because it regulates metabolism, is used in drug discovery and helps identify disease conditions.

Enzyme Classification

LESSON 12

1. Classification of Enzymes

Enzymes are classified on the basis of the reaction they catalyse into six groups: oxido-reductases, transferases, hydrolases, lyases, isomerases and ligases.

- **Oxido-reductases:** Carry out oxidation-reduction reactions.
- **Transferases:** Transfer a group.
- **Hydrolases:** Break bonds by adding water.
- **Lyases:** Break bonds without hydrolysis.
- **Isomerases:** Rearrange atoms within a molecule.
- **Ligases:** Join molecules in condensation reactions.

2. Classification by Substrate

Enzymes are also classified by substrate, according to the substance they act on.

Enzyme Type	Substrate
Proteases	Proteins
Lipases	Fats
Glycosidases	Carbohydrates
Nucleases	Nucleic acids

This classification shows the specific nature of enzymes: each group is named for the type of reaction it catalyses or for the substrate on which it acts.

3. Summary of the Two Bases

On the basis of reaction, oxido-reductases carry out oxidation-reduction, transferases transfer a group, hydrolases break bonds by adding water, lyases break bonds without hydrolysis, isomerases rearrange atoms within a molecule, and ligases join molecules in condensation reactions. On the basis of substrate, proteases act on proteins, lipases act on fats, glycosidases act on carbohydrates and nucleases act on nucleic acids.

Cell Structure and Function

UNIT 3

Techniques Used in Cell Biology

LESSON 13

1. Cell Fractionation

A primary objective of **cell fractionation** is to separate the major organelles so that their particular functions can be determined. By separating the major organelles, the particular function of each organelle can be studied.

2. Resolution

Resolution is the clarity of an image. It is the ability to distinguish two close points as separate.

- **Cell fractionation:** separates the major organelles so that the particular function of each organelle can be determined and studied.
- **Resolution:** the clarity of an image — the ability to distinguish two close points as separate.

Cell Wall and Plasma Membrane

LESSON 14

1. Why the Plant Cell Wall Is Rigid

The plant cell wall is rigid because it is mainly made of **cellulose** microfibrils embedded in a matrix of pectin and other polysaccharides. Cellulose chains are held together by hydrogen bonds, forming strong bundles that resist stretching. In many cells lignin is also deposited, which makes the wall even harder. This rigidity gives shape, strength and protection to the plant cell.

2. Structure and Composition of the Cell Wall

The plant cell wall lies outside the plasma membrane and is mainly composed of cellulose, pectin and other polysaccharides, with lignin in woody cells.

- **Primary wall:** thin and flexible, formed first.
- **Secondary wall:** thicker and more rigid, deposited later.
- **Middle lamella:** made of pectin; it joins adjacent cells.

The wall gives shape, mechanical strength and protection to the cell, prevents it from bursting, and still allows water and solutes to pass through it.

3. Why the Plasma Membrane Is Differentially Permeable

The plasma membrane is differentially (selectively) permeable because its phospholipid bilayer allows only certain substances to pass. Small non-polar molecules pass easily, while ions and large polar molecules need specific transport proteins. This selective passage is due to the hydrophobic core of the bilayer and the presence of channel and carrier proteins. It helps the cell maintain its internal environment.

4. Chemical Composition of the Plasma Membrane

The plasma membrane is mainly composed of a phospholipid bilayer with proteins embedded in it, along with cholesterol and small amounts of carbohydrates. This fluid mosaic structure makes the membrane flexible and selectively permeable.

5. Functions of the Plasma Membrane

- **Regulating entry and exit:** it regulates the entry and exit of ions, nutrients and waste products through diffusion, osmosis and active transport.
- **Receiving signals:** membrane receptors receive signals from the environment and from other cells.
- **Cell recognition:** glycoproteins help in cell recognition.

In this way the plasma membrane controls the cell's interactions with its environment.

Organelles I — Endoplasmic Reticulum, Golgi Complex, Lysosomes and Vacuoles

LESSON 15

1. The Cytoplasm

The cytoplasm is shown in both the animal cell and the plant cell. In plant cells, a large central vacuole reduces the volume of the cytoplasm. When a lysosome ruptures, its enzymes are released into the cytoplasm and digest the cell's own components. In prokaryotic cells, the ribosomes are free in the cytoplasm.

2. Rough and Smooth Endoplasmic Reticulum

The endoplasmic reticulum is a network of tube-like channels of lipoprotein extending from the plasma membrane to the nuclear membrane. Both types transport materials within the cell and provide a surface for biochemical reactions.

Feature	Rough Endoplasmic Reticulum	Smooth Endoplasmic Reticulum
Surface	Has ribosomes attached to its surface, giving it a rough appearance	Has no ribosomes on its surface
Functions	Takes part in protein synthesis, folding and transport of proteins	Involved in lipid and steroid synthesis, detoxification and storage of calcium ions

3. Golgi Complex — Structure and Functions

The Golgi complex consists of flattened membrane-bound sacs called **cisternae**, stacked together, with small vesicles around them. It is especially prominent in glandular cells.

- **Processing:** products of the endoplasmic reticulum reach the Golgi complex, where they are modified, sorted, stored and packed into vesicles.
- **Delivery:** these vesicles then send the products to their destination inside or outside the cell.
- **Forming lysosomes:** the Golgi complex also forms lysosomes.
- **Secretion:** it secretes substances such as hormones, enzymes and mucus.

4. Lysosomes — Compartmentalisation of Enzymes

In animal cells, hydrolytic enzymes are packed in lysosomes to prevent general destruction of cellular components. Lysosomes pack hydrolytic enzymes in a membrane-bound sac so they digest only targeted material and do not destroy the whole cell.

In **Tay-Sachs disease**, a lysosomal enzyme is missing, so complex lipids cannot be broken down and accumulate in cells, which become clogged with very large and complex lipids.

5. Lysosomes — Structure

Lysosomes are small spherical sacs bounded by a single membrane and filled with hydrolytic enzymes. They are formed from the Golgi complex, and a newly formed lysosome before it starts its function is called a **primary lysosome**. Their enzymes work best at acidic pH inside the lysosome, which protects the rest of the cell.

6. Lysosomes — Functions

- **Digestion:** lysosomes digest food particles, worn-out organelles and invading bacteria and viruses.
- **Autophagy:** they also carry out autophagy.
- **Autolysis:** when they rupture, they cause autolysis of the cell.

7. Why Lysosomes Are Called Suicidal Sacs

Lysosomes are called suicidal sacs because they contain powerful hydrolytic enzymes capable of digesting the whole cell. When a cell is damaged or old, the lysosomal membrane may rupture and release these enzymes into the cytoplasm. The enzymes then digest the cell's own components, causing autolysis. This self-destruction is useful for removing worn-out cells and organelles.

8. Vacuoles

Plant cells contain a large central vacuole that occupies most of the volume enclosed by the plasma membrane. The volume enclosed by the plasma membrane of plant cells is often much larger than the corresponding volume in animal cells, because plant cells contain a large vacuole that reduces the volume of the cytoplasm and makes plant cells much larger in volume. A large central vacuole is a plant cell feature.

9. Peroxisomes and Glyoxysomes

Peroxisomes are mainly concerned with the detoxification of alcohol and the breakdown of hydrogen peroxide.

Feature	Peroxisome	Glyoxysome
Where found	Microbodies found in both plant and animal cells	Specialised peroxisomes found only in plant cells, especially in germinating seeds
Main work	Mainly concerned with detoxification of alcohol and breakdown of hydrogen peroxide by the enzyme catalase	Contain enzymes that convert stored fatty acids into sugars needed for seedling growth
Special role	Work generally in detoxification	Have a special role in seed germination

Organelles II — Mitochondria, Chloroplasts, Ribosomes, Cytoskeleton and Nucleus

LESSON 16

1. Mitochondria — Energy Transformers of the Cell

Mitochondria are one of the main energy transformers of cells. They transform energy from food into ATP, the usable energy form of the cell.

2. Why Mitochondria Are Called the Power House of the Cell

Mitochondria are called the power house of the cell because they produce most of the cell's ATP during aerobic respiration. Glucose and other fuels are oxidised in the Krebs cycle and electron transport chain located in mitochondria. The energy released is stored in ATP, which powers all cell activities. Without mitochondria the cell would not get enough usable energy.

3. Structure of Mitochondria

Mitochondria are rod-shaped organelles bounded by a double membrane.

- **Outer membrane:** smooth.
- **Inner membrane:** folded into **cristae** that increase its surface area. The cristae carry the electron transport chain and ATP synthase for oxidative phosphorylation.
- **Matrix:** the space inside the inner membrane, which contains enzymes, circular DNA and 70S ribosomes.

Because of their own DNA and ribosomes, mitochondria have a semi-autonomous existence in the cell. In a diagram of a mitochondrion, the outer membrane, the inner membrane/cristae and the matrix are labelled.

4. How Mitochondria Are Similar to Bacteria

- **DNA and ribosomes:** they have their own circular DNA and 70S ribosomes like bacteria, and they can make some of their own proteins.
- **Double membrane:** they are surrounded by a double membrane.
- **Division:** they divide by a process similar to binary fission.
- **Size:** their size is also similar to that of bacteria.

This supports the idea that mitochondria evolved from ancient bacteria.

5. Mitochondria in Plant and Animal Cells

Mitochondria are present in both plant and animal cells. They are a structure common to plant and animal cells.

6. Chloroplasts — Energy Converting Organelle

The chloroplast is called an energy converting organelle because it converts light energy into chemical energy. During photosynthesis, chlorophyll absorbs sunlight and uses it to make ATP and $NADPH_2$, which then fix carbon dioxide into glucose. In this way solar energy is changed into stored chemical energy of food. No other organelle performs this conversion in plants.

7. Organelles That Contain DNA

Besides the nucleus, only chloroplasts and mitochondria contain their own DNA; ribosomes do not.

8. Ribosomes and Protein Synthesis

Ribosomes are the sites of protein synthesis, so protein-producing cells are rich in ribosomes. Large numbers of ribosomes are present in cells that specialise in producing proteins.

9. Cytoskeleton — Types, Structure, Composition and Functions

The cytoskeleton is a network of protein fibres that gives the cell its three-dimensional shape. It constantly reorganises itself according to the needs of the cell, and it also helps in intracellular transport of organelles and vesicles.

Type	Made of	Functions
Microfilaments	Actin	Help in cell movement and muscle contraction
Intermediate filaments	—	Provide mechanical strength
Microtubules	Tubulin protein	Form cilia, flagella, centrioles and the spindle during cell division

10. The Nucleus

The nucleus is present in both the animal cell and the plant cell. The nucleus contains DNA; besides the nucleus, only chloroplasts and mitochondria contain their own DNA. The endoplasmic reticulum extends from the plasma membrane to the nuclear membrane.

Prokaryotic and Eukaryotic Cells

LESSON 17

1. Prokaryotic and Eukaryotic Ribosomes

Feature	Prokaryotic Ribosome	Eukaryotic Ribosome
Size / type	Smaller, of 70S type	Larger, of 80S type
Subunits	Made of 50S and 30S subunits	Made of 60S and 40S subunits
Position	Free in the cytoplasm	May be free or attached to the endoplasmic reticulum

This difference is also important in medicine, because some antibiotics target only bacterial 70S ribosomes.

2. Mitochondria and the Bacterial Comparison

Mitochondria of eukaryotic cells resemble bacteria in several ways: they have their own circular DNA and 70S ribosomes like bacteria, they are surrounded by a double membrane, they divide by a process similar to binary fission, and their size is similar to that of bacteria. This supports the idea that mitochondria evolved from ancient bacteria.

3. Plant Cells and Animal Cells

Mitochondria are common to plant and animal cells. Chloroplast, cell wall and a large central vacuole are plant features.

The animal cell shows cytoplasm, cell membrane, nucleus, mitochondrion, ribosomes and vacuole. The plant cell shows cell wall, chloroplast, vacuole, nucleus and cytoplasm.

Structure	Plant Cell	Animal Cell
Mitochondria	Present	Present — common to both
Chloroplast	Present	A plant feature
Cell wall	Present	A plant feature
Central vacuole	Large central vacuole present	A plant feature

The volume enclosed by the plasma membrane of plant cells is often much larger than the corresponding volume in animal cells, because plant cells contain a large central vacuole.

Bioenergetics

UNIT 4

Photosynthesis — Light Reactions

LESSON 18

1. Photosystems and Pigments

Chlorophyll-a is almost identical to chlorophyll-b, but the small structural difference between them is enough to give them a different absorptive (absorption) spectrum. In a photosystem, chlorophyll is organized along with other molecules, and chlorophyll and other pigments form the light-gathering **antenna complex** that passes energy to the reaction centre.

2. Why the Antenna Complex Contains Other Pigments

The antenna complex contains other pigments along with chlorophyll so that a wider range of light wavelengths can be absorbed. Accessory pigments like carotenoids and chlorophyll-b absorb light of different colours which chlorophyll-a alone cannot absorb well. They pass the absorbed energy to the reaction centre. This makes photosynthesis more efficient, especially in varying light conditions. Photosystem I is located in the non-appressed (non-stacked) regions of the thylakoid membrane, along with ATP synthase and NADP reductase.

3. Photosynthesis as a Redox Process

Photosynthesis is called a redox process because it involves both oxidation and reduction. Water is oxidized, losing electrons and releasing oxygen. Carbon dioxide is reduced, gaining electrons and hydrogen, and is converted into glucose. Since oxidation and reduction occur together in the same process, photosynthesis is a redox process.

4. Cyclic and Non-Cyclic Photophosphorylation

Feature	Non-Cyclic Photophosphorylation	Cyclic Photophosphorylation
Photosystems	Both Photosystem II and Photosystem I work in series	Only Photosystem I takes part
Electron flow	Linear flow of electrons from water to NADP+	Electrons are passed back to Photosystem I through carriers
Products	Oxygen is released, and both ATP and NADPH2 are produced	No oxygen is evolved and no NADPH2 is made, only ATP is formed

Both processes use light energy to make ATP by chemiosmosis across the thylakoid membrane.

5. How Cyclic Photophosphorylation Helps

In cyclic photophosphorylation, electrons from Photosystem I return to it through electron carriers, and only ATP is produced. This extra ATP is needed because the Calvin cycle requires more ATP than NADPH2. Cyclic flow therefore balances the supply of ATP and NADPH2 for carbon fixation. It also protects the plant when NADP+ is not available.

Photosynthesis — Light-Independent Reactions and Alternative CO2 Fixation

LESSON 19

1. The Calvin Cycle — Light-Independent Phase

The light independent phase, or Calvin cycle, takes place in the stroma of the chloroplast and does not directly need light. This phase depends on the products of the light phase but itself is light independent. The sequence is: CO2 + RuBP (5C) → PGA (3C) → glyceraldehyde phosphate → glucose and other carbohydrates, with regeneration of RuBP.

- **Carboxylation:** CO2 is fixed by RuBP (5C) with the enzyme Rubisco to form PGA (3C). Carboxylation in the Calvin cycle is carried out by the enzyme Rubisco carboxylase, which fixes CO2 onto RuBP.

- **Reduction:** PGA is reduced to glyceraldehyde phosphate using ATP and NADPH2 produced in the light reactions.
- **Regeneration:** Some of this product is used to make glucose and other carbohydrates, while the rest regenerates RuBP so the cycle can continue.

2. Why the Calvin Cycle is Called the C3 Cycle

The Calvin cycle is also called the C3 cycle because the first stable product of carbon fixation in it is a three-carbon compound. Carbon dioxide is fixed by RuBP with the help of Rubisco to form phosphoglyceric acid (PGA), which contains three carbon atoms. Plants using only this cycle are called C3 plants. Most common plants, such as wheat and rice, are C3 plants.

3. Alternative CO2 Fixation — C4 Plants

Some plants avoid photorespiration by alternative CO2 fixation mechanisms. In C4 plants like sugarcane and maize, CO2 is first fixed in mesophyll cells by PEP into oxaloacetate (4C), which is converted to malate and transported to bundle sheath cells, where CO2 is released for the Calvin cycle. In members of the grass family, oxaloacetic acid is the first / intermediate carbon-fixing compound that passes CO2 to the Calvin cycle.

4. Alternative CO2 Fixation — CAM Plants

In CAM plants like cacti and pineapple, stomata open at night and CO2 is stored as organic acids in vacuoles, then released in the day for fixation. CAM plants close their stomata during the day to conserve water in hot and dry conditions. They open stomata at night, when evaporation is low, and fix CO2 into organic acids stored in vacuoles. During the day these acids release CO2 internally for the Calvin cycle while stomata remain closed. This greatly reduces transpiration and helps succulent plants survive in deserts. Both C4 and CAM mechanisms keep CO2 concentration high around Rubisco and conserve water or increase efficiency.

Cellular Respiration — Glycolysis and the Krebs Cycle

LESSON 20

1. Cellular Respiration

Cellular respiration is the breakdown of glucose by redox reactions to synthesize ATP, releasing energy for the cell. Aerobic respiration occurs in the presence of oxygen and completely oxidizes glucose to CO2 and water, producing a large amount of ATP through glycolysis, Krebs cycle and electron transport chain.

2. Glycolysis — Breaking of Glucose in the Cytosol

The breaking of glucose in the cytosol is glycolysis, an anaerobic process common to almost all cells. Glycolysis does not require oxygen and takes place entirely in the cytosol. The sequence is: glucose (6C) → phosphorylated glucose → two molecules of pyruvate (3C).

- **Phosphorylation:** One glucose molecule (6C) is first phosphorylated using ATP and then split into two molecules of pyruvate (3C).
- **Net gain:** A net gain of 2 ATP by substrate-level phosphorylation and 2 NADH2 molecules is obtained.
- **Fate of pyruvate:** The pyruvate formed then enters mitochondria for aerobic respiration or is fermented in anaerobic conditions.

Oxidation of one pyruvate gives one acetyl CoA with 2 carbon atoms, which are fed into the citric acid cycle.

3. The Tricarboxylic Acid (Krebs) Cycle

The tricarboxylic acid (Krebs) cycle takes place in the mitochondrial matrix. The sequence is: Acetyl CoA (2C) + oxaloacetate (4C) → citrate (6C) → oxaloacetate, releasing 2 CO2. Acetyl CoA (2C) combines with oxaloacetate (4C) to form citrate (6C), the first tricarboxylic acid. Through a series of steps involving oxidative decarboxylation, citrate is converted back to oxaloacetate, releasing 2 CO2. Oxidative decarboxylation of isocitrate in the Krebs cycle forms α-ketoglutarate, releasing CO2.

- **Per turn:** 3 NADH2, 1 FADH2 and 1 ATP (by substrate-level phosphorylation) are produced.
- **Per glucose:** Since one glucose gives two turns, the yield is doubled, and the reduced coenzymes then feed the electron transport chain.
- **Substrate-level phosphorylation:** One glucose gives two turns of the Krebs cycle, and each turn forms one ATP by substrate-level phosphorylation, so two ATP are

formed.

Cellular Respiration — Electron Transport Chain, Chemiosmosis and Fermentation

LESSON 21

1. ATP — Common Energy Currency

ATP is called the common energy currency of living systems because it stores and supplies energy for almost all cellular activities. Energy from respiration and photosynthesis is trapped in ATP and later released by its hydrolysis to ADP wherever work is done. All organisms, from bacteria to humans, use this same molecule for energy transfer. Just like money, it is earned, stored for a short time and spent on needs.

2. Electron Transport Chain and the Role of Oxygen

Oxygen is consumed directly as the final electron acceptor at the end of the electron transport chain, forming water.

3. Chemiosmosis and Oxidative Phosphorylation

Chemiosmosis is the process in which ATP is synthesized using energy of a proton gradient across a membrane. The sequence is: NADH2 and FADH2 → electrons pass through carriers → H+ ions are pumped, creating a gradient → protons flow back through ATP synthase → ADP and phosphate join to form ATP. Protons flow back through ATP synthase, and this flow drives the joining of ADP and phosphate to form ATP. Oxidative phosphorylation is this ATP formation linked with the oxidation of reduced coenzymes, with oxygen as the final electron acceptor. It produces most of the ATP of aerobic respiration in mitochondria.

4. Aerobic Respiration and Fermentation

Feature	Aerobic Respiration	Anaerobic Respiration / Fermentation
Oxygen	Occurs in the presence of oxygen	Occurs without oxygen
Breakdown	Completely oxidizes glucose to CO2 and water	Glucose is only partially broken down
ATP	Produces a large amount of ATP	Only 2 ATP per glucose

Anaerobic respiration types include alcoholic fermentation, producing ethanol and CO2 in yeast, and lactic acid fermentation, producing lactic acid in muscles and some bacteria, with only 2 ATP per glucose.

5. Why Oxidation of Pyruvate Gives More Energy than Lactic Acid Fermentation

Oxidation of pyruvate in aerobic respiration completely breaks it down to CO2 and water through the Krebs cycle and electron transport chain. This releases a large number of ATP molecules from one glucose molecule. In lactic acid fermentation, glucose is only partly broken down and lactic acid still contains much unused energy. Therefore aerobic oxidation of pyruvate provides far more energy than fermentation.

Photorespiration

LESSON 22

1. Photorespiration and the Need to Avoid It

Some plants avoid photorespiration by alternative CO2 fixation mechanisms. Both mechanisms keep CO2 concentration high around Rubisco and conserve water or increase efficiency. These alternative mechanisms are found in C4 plants and in CAM plants, in contrast to plants that use only the Calvin cycle.

2. Rubisco and Carboxylation

Carboxylation in the Calvin cycle is carried out by the enzyme Rubisco carboxylase, which fixes CO2 onto RuBP. In the Calvin cycle, CO2 is fixed by RuBP (5C) with the enzyme Rubisco to form PGA (3C), a three-carbon compound. Plants using only this cycle are called C3 plants, and most common plants, such as wheat and rice, are C3 plants. Keeping CO2 concentration high around Rubisco is the shared effect of the alternative mechanisms described below.

3. C4 Plants — Spatial Separation

In C4 plants like sugarcane and maize, CO2 is first fixed in mesophyll cells by PEP into oxaloacetate (4C), which is converted to malate and transported to bundle sheath cells, where CO2 is released for the Calvin cycle. In members of the grass family, oxaloacetic acid is the first / intermediate carbon-fixing compound that passes CO2 to the Calvin cycle. In this way CO2 is delivered to the Calvin cycle in bundle sheath cells while its concentration is kept high around Rubisco.

4. CAM Plants — Temporal Separation

In CAM plants like cacti and pineapple, stomata open at night and CO2 is stored as organic acids in vacuoles, then released in the day for fixation. CAM plants close their stomata during the day to conserve water in hot and dry conditions. They open stomata at night, when evaporation is low, and fix CO2 into organic acids stored in vacuoles. During the day these acids release CO2 internally for the Calvin cycle while stomata remain closed. This adaptation greatly reduces transpiration and helps succulent plants survive in deserts, while also keeping CO2 available for fixation during the day.

Acellular Life

UNIT 5

Viruses — Discovery and Structure

LESSON 23

1. Structure of a Virus

A virus particle consists of a **Protein coat (capsid)**, **Spike (attachment protein)** and **Nucleic acid (DNA or RNA)**. The nucleic acid is either DNA or RNA, protected inside the capsid.

2. Living and Non-living Status

Viruses show both living and non-living characters. Outside a host they are inert particles that can be crystallized, have no metabolism and do not respire or grow, which are non-living characters. Inside a living host cell they reproduce, mutate and evolve using the host machinery, which are living characters. Because of this dual nature, viruses are placed at the boundary between living and non-living and are called acellular particles.

3. Survival Without a Host

Outside a host, a virus survives as an inert particle called a **virion**, protected by its protein capsid and sometimes an envelope. It shows no metabolism and can remain inactive for long periods on surfaces, in air, water or soil, depending on temperature and humidity. Some viruses survive unfavourable conditions in dormant form inside host cells or seeds. When a suitable host is contacted, the virus becomes active again and starts its infection cycle.

4. Positive and Negative Sense Viruses

A **positive sense virus** has single-stranded RNA that can act directly as messenger RNA and be translated into proteins by the host ribosomes. A **negative sense virus** has RNA complementary to mRNA, so it must first be transcribed into a positive strand by viral RNA polymerase before proteins can be made. This classification is based on the nature of the viral genome. Influenza virus is a negative sense virus, while polio virus is positive sense.

5. Baltimore Classification

The Baltimore classification divides viruses into seven groups according to their genome and method of mRNA synthesis. This system is widely used for virus classification.

- **Group I:** double-stranded DNA viruses
- **Group II:** single-stranded DNA viruses
- **Group III:** double-stranded RNA viruses
- **Group IV:** positive sense single-stranded RNA viruses
- **Group V:** negative sense single-stranded RNA viruses
- **Group VI:** RNA retroviruses that use reverse transcriptase
- **Group VII:** double-stranded DNA retroviruses

Parasitic Nature of Viruses

LESSON 24

1. Obligate Parasitic Nature

Viruses have no machinery of their own, so they use the host cell's organelles, enzymes and energy for their reproduction and evolution. Viruses evolve by using the cellular organelles, cellular enzymes and cellular energy — all of them. Outside a host they remain inert, and inside a living host cell they reproduce, mutate and evolve using the host machinery.

2. How a Virus Takes Over the Host Cell

Viruses cause diseases by penetrating host cells, multiplying inside them and damaging tissues. The pathogenicity of a virus depends upon the effective penetration of its genome into the host cell, after which it takes over the cell machinery. Pathogenicity also depends on host immunity and environmental conditions, including the immunity of the host body and the overall environment inside the body.

3. Formation of New Virus Particles

In **assembly**, the smaller proteins are cut down and viral proteins are cut/processed and combined with the viral genome to form new complete virus particles. This is the process by which a new virus structure is formed inside the host cell using host resources.

4. Dependence on the Host

- **No own machinery:** viruses use host organelles, enzymes and energy.
- **No metabolism outside host:** as a virion, the virus shows no metabolism and does not respire or grow.
- **Activity only in host:** reproduction, mutation and evolution occur only inside a living host cell.
- **Survival as virion:** protected by capsid and sometimes an envelope, inactive on surfaces, in air, water or soil until a suitable host is contacted.

Because of this complete dependence, viruses are described as acellular particles at the boundary between living and non-living.

Life Cycle of Bacteriophage

LESSON 25

1. How Bacteriophage Infects Bacteria

A bacteriophage first attaches to specific receptors on the bacterial surface by its tail fibres. Its tail sheath contracts and the viral DNA is injected into the bacterium through the cell wall, while the protein coat stays outside. The viral DNA then takes over the bacterial machinery to make new phage parts. Finally the cell bursts and releases new phages, or the viral DNA may stay integrated in a lysogenic cycle.

2. Lytic Life Cycle

Attachment → Penetration (injection of DNA) → Synthesis of phage components → Assembly of new phages → Release by lysis of the host cell.

In detail: the bacteriophage attaches to the bacterial surface with its tail fibres → the tail contracts and viral DNA is injected into the cell, leaving the coat outside → the phage DNA takes over host machinery, host DNA is broken down and new phage DNA and proteins are synthesized → these parts are assembled into complete phages, and the enzyme **lysozyme** weakens the cell wall → the bacterium bursts (lysis) and releases a large number of new phages to infect other cells. Lysozyme made by the phage dissolves the bacterial cell wall, so new bacteriophages escape from the host cell.

3. Lysogenic Life Cycle

Attachment → Penetration → Integration of phage DNA as prophage into the bacterial chromosome → Replication of prophage with the host DNA during cell division. Under induction, the prophage may leave the chromosome and enter the lytic cycle. Both cycles begin the same way but differ after penetration.

4. Lytic vs Lysogenic Cycle

Feature	Lytic Cycle	Lysogenic Cycle
Action of phage DNA	Immediately takes control of the bacterium	Integrates into bacterial chromosome as a prophage
New phages	Many new phages are produced at once	Prophage is copied silently with host DNA for many generations
Fate of host	Host cell is lysed (burst) to release phages, so the host dies	Initially harmless; host survives
Later course	Destructive at once	May later switch to the lytic cycle under certain conditions

Life Cycle of HIV

LESSON 26

1. Reverse Transcription in HIV

Reverse transcription is the formation of DNA from an RNA template, the reverse of normal transcription. HIV carries the enzyme **reverse transcriptase**, which converts its single-stranded RNA into double-stranded DNA inside the host cell. This viral DNA enters the nucleus and integrates into the human chromosome as provirus. It then directs the production of new HIV particles, mainly in helper T lymphocytes.

2. Life Cycle of HIV

HIV attaches to CD4 receptors of helper T lymphocytes → HIV enters the cell → Its RNA is converted into double-stranded DNA by reverse transcriptase → This DNA integrates into the host chromosome as provirus → The provirus may remain latent or become active, producing viral RNA and proteins using host machinery → New HIV particles are assembled at the cell membrane and released by budding, acquiring an envelope → Gradually the helper T cells are destroyed, the immune system weakens, and AIDS develops.

3. Symptoms of AIDS

AIDS is the final stage of HIV infection in which the immune system is badly damaged. Its symptoms include prolonged fever, night sweats, rapid weight loss, chronic diarrhoea and swollen lymph glands. The patient suffers repeated opportunistic infections such as pneumonia and tuberculosis, and skin lesions may appear. Because helper T cells are destroyed, even common infections become severe and life-threatening.

4. Prevention and Control of AIDS

- **Blood safety:** avoid contact with infected blood and body fluids; blood must be screened before transfusion.
- **Sterile equipment:** sterilized or disposable syringes should be used; avoid sharing of needles and razors.
- **Safe practices:** safe sexual practices are essential.
- **Mother to baby:** an infected mother can reduce transmission to her baby by proper medical treatment during pregnancy and delivery.

There is no complete cure, so prevention and awareness are the main methods of control.

Viral Diseases

LESSON 27

1. Viral Diseases in Humans

Viruses cause diseases by penetrating host cells, multiplying inside them and damaging tissues. Human diseases like influenza, hepatitis, dengue, polio, measles and AIDS cause millions of deaths and huge medical expenses. Poliovirus damages motor neurons of the spinal cord and causes paralysis, the disease poliomyelitis. Dengue fever, encephalitis and yellow fever are caused by **arbo-viruses**, which are transmitted by arthropods such as mosquitoes. Aedes mosquito is the vector that transmits dengue virus (and also chikungunya virus) through blood.

2. Animal and Plant Viruses

Virus	Vector / Transmission	Symptoms / Effects
Dengue virus	Aedes mosquito through blood	Fever, rash and pain behind the eyes
Hepatitis C virus	Infected blood	Fever, yellow skin and fatigue
Measles virus	Airborne droplets	Fever, cough and body rash
Tobacco mosaic virus	Contact	Mosaic patterns on leaves
Cotton leaf curl virus	Whitefly	Curling of leaves and great loss of cotton yield

Each virus has a specific vector, mode of transmission and symptoms.

3. Pathogenicity and Economic Losses

Pathogenicity depends on host immunity and environmental conditions. Epidemics reduce working capacity, close schools and industries, and burden hospitals, as seen during COVID-19 when production fell in most countries. Plant viruses such as cotton leaf curl virus have caused losses of billions of rupees to Pakistan's cotton economy. Thus viruses damage both human health and the national economy.

Prions and Viroids

LESSON 28

1. Prions

Prions are infectious protein particles that cause pathogenicity in humans and animals, e.g. mad cow disease. They are proteins that cause diseases in humans and animals. Prions are infectious particles made only of protein, with no nucleic acid, and they mainly cause diseases of the nervous system in humans and animals. Prions were discovered by Stanley Prusiner.

2. Viroids

Viroids are subviral particles of single-stranded RNA with some double-stranded regions, smaller than viruses and without a protein coat. They are smaller than viruses having single stranded RNA with some double stranded regions. Viroids are infectious particles made only of small single-stranded RNA, with no protein coat, and they cause diseases only in plants. Viroids were discovered by Theodor Diener.

3. Prions vs Viroids

Feature	Prions	Viroids
Chemical nature	Only protein, no nucleic acid	Only small single-stranded RNA, no protein coat
Hosts	Humans and animals	Plants only
Diseases	Diseases of the nervous system, e.g. mad cow disease	Diseases only in plants
Discovered by	Stanley Prusiner	Theodor Diener

4. Virus vs Prion vs Viroid

Feature	Virus	Prion	Viroid
Composition	Protein coat (capsid) + nucleic acid (DNA or RNA)	Only protein	Only single-stranded RNA
Protein coat	Present (capsid)	Particle itself is protein	No protein coat
Hosts / Diseases	Humans, animals and plants	Humans and animals — nervous system diseases	Plants only

Both prions and viroids are subviral particles, simpler and smaller than viruses, but their chemical nature and hosts are different.

Prokaryotes

UNIT 6

Taxonomy of Prokaryotes and Archaea

LESSON 29

1. Three Domains of Classification

Living organisms are classified into three domains: Bacteria, Archaea and Eukarya. Bacteria are true prokaryotes with peptidoglycan in their cell walls and are found everywhere. Archaea are also prokaryotic, but their cell walls lack peptidoglycan, their membrane lipids are different, and many live in extreme environments like hot springs and salty lakes. Eukarya includes all eukaryotic organisms — protoctists, fungi, plants and animals — whose cells have a true nucleus and membrane-bound organelles. The domains differ in cell structure, biochemistry and habitat.

Feature	Bacteria	Archaea	Eukarya
Cell type	Prokaryotic	Prokaryotic	Eukaryotic, with a true nucleus and membrane-bound organelles
Cell wall	Peptidoglycan present	Peptidoglycan absent	—
Membrane lipids	—	Different from bacteria	—
Habitat / members	Found everywhere	Many in extreme environments like hot springs and salty lakes	Protoctists, fungi, plants and animals

2. Protobionts and Their Relationship with Prokaryotes

Protobionts were the first non-cellular structures formed from organic molecules on the early Earth, considered precursors of living cells. They could absorb substances, grow, and show simple metabolism and division, but they could not reproduce exactly like true cells. Over time, protobionts that developed a nucleic acid system and a membrane gave rise to the first true prokaryotic cells. Prokaryotes are therefore regarded as the evolved, fully cellular descendants of protobionts. This relationship explains the origin of the earliest bacteria-like organisms.

3. Archaea in Extreme Conditions — Extremophiles

Archaea living in extreme conditions are called extremophiles. These archaea survive where most other organisms cannot live.

- **Thermophiles:** archaea living in very hot places, such as hot springs.
- **Thermoacidophiles:** archaea tolerating both heat and acidity.
- **Halophiles:** archaea living in highly salty environments like salt lakes.
- **Methanogens:** archaea that live in oxygen-free places such as marshes and animal guts and produce methane gas.

Bacteria — Ecology and Diversity

LESSON 30

1. Bacteria in Nature

Bacteria are true prokaryotes and are found everywhere. They live in soil, air and water, as decomposers of dead matter, as nitrogen-fixing organisms in soil and in root nodules, and normally on and inside the human body.

2. Bacteria as Recycling Agents in Nature

Bacteria are called recycling agents because decomposer bacteria break down dead bodies of plants and animals and organic waste into simple inorganic substances. These substances, such as CO_2, water, ammonia and minerals, return to the soil, air and water and are used again by plants. Without bacteria, dead matter would pile up and nutrients would remain locked in it. Bacteria also play a key role in nitrogen, carbon and sulphur cycles, keeping nature's recycling continuous.

3. How Bacteria Increase Soil Fertility

Bacteria increase soil fertility mainly by fixing atmospheric nitrogen and decomposing organic matter.

- **Rhizobium:** living in root nodules of leguminous plants, it converts nitrogen gas into ammonia compounds usable by plants.

- **Azotobacter:** a free-living nitrogen-fixing bacterium in soil.
- **Nostoc:** a cyanobacterium that also fixes nitrogen in soil.
- **Decomposer bacteria:** they convert dead remains into humus and release nitrates through nitrification.

As a result the soil becomes rich in plant nutrients and crop yield increases.

4. Normal Flora of Bacteria

Normal flora means the microorganisms, mainly bacteria, that normally live on the skin and mucous membranes and inside the intestine of a healthy person without causing disease. The human body contains about one hundred trillion such bacteria. They prevent colonization by harmful pathogens by competing for attachment sites and nutrients. They also stimulate antibody production and some synthesize vitamins K and B12 in the large intestine. Internal tissues like blood and muscles are normally free of these microbes.

Structure, Shape and Size of Bacteria

LESSON 31

1. Structure of a Bacterium

A bacterium is a unicellular prokaryote surrounded by a cell membrane and a cell wall of peptidoglycan, sometimes with an outer capsule. Its cytoplasm contains a single circular DNA in the nucleoid region, plasmids, 70S ribosomes and storage granules, but no membrane-bound organelles. Many bacteria have flagella for movement, and pili or fimbriae for attachment and conjugation. Some form resistant endospores in unfavourable conditions.

The labelled diagram of a bacterial cell in the textbook shows: capsule, cell wall, membrane, nucleoid (DNA), plasmid, ribosome and flagellum.

- **Capsule:** an outer covering present in some bacteria, outside the cell wall.
- **Cell wall:** made of peptidoglycan.
- **Nucleoid:** the region containing a single circular DNA.
- **Plasmids:** present in the cytoplasm along with 70S ribosomes and storage granules.
- **Flagella:** for movement.
- **Pili or fimbriae:** for attachment and conjugation.
- **Endospores:** resistant structures formed by some bacteria in unfavourable conditions.

2. Structure of Peptidoglycan

Peptidoglycan is a large polymer that forms the rigid framework of the bacterial cell wall. It consists of long chains of two alternating sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). Short peptide chains are attached to NAM units and cross-link the sugar chains together, forming a strong mesh-like network. This structure gives the wall strength, maintains cell shape and protects the bacterium from bursting. The enzyme lysozyme and the antibiotic penicillin attack this layer.

3. Gram-Positive and Gram-Negative Bacteria

Feature	Gram-positive bacteria	Gram-negative bacteria
Peptidoglycan layer	Thick	Thin
Outer membrane	No outer membrane	An additional outer membrane containing lipopolysaccharides
Gram staining	Retain the crystal violet stain and appear purple	Lose the violet stain and take the pink counter stain
Resistance to antibiotics	—	Generally more resistant, because of the outer membrane
Example	Staphylococcus	Escherichia coli

Modes of Nutrition in Bacteria

LESSON 32

1. Autotrophic Mode of Nutrition

Autotrophic bacteria make their own food from simple inorganic substances. They do not depend on other organisms for food.

- **Photoautotrophic bacteria:** such as cyanobacteria and green and purple sulphur bacteria, they use sunlight as an energy source for photosynthesis.
- **Chemoautotrophic bacteria:** they obtain energy by oxidizing inorganic substances like ammonia, nitrites, sulphur or iron, and use it to fix CO_2.

These bacteria are important in nutrient cycles, e.g. nitrifying bacteria in soil.

2. Photosynthesis Mechanism in Bacteria

Bacterial photosynthesis uses light energy to make food, but most bacteria use bacteriochlorophyll instead of chlorophyll and do not release oxygen. The light energy is trapped to form ATP and reducing power, which fix CO_2 into organic compounds.

- **Anoxygenic photosynthesis:** in purple and green sulphur bacteria, hydrogen sulphide acts as the hydrogen donor, so sulphur is deposited instead of oxygen being evolved.
- **Cyanobacteria:** an exception — they use chlorophyll-a and water as donor and release oxygen, like plants.

Growth and Reproduction in Bacteria

LESSON 33

1. Bacterial Growth

Bacterial growth means an increase in the number of bacterial cells by binary fission. When bacteria are grown in a culture, growth shows four phases.

2. Phases of Bacterial Growth

- **Lag phase:** cells adjust to the medium and grow in size but do not divide much.
- **Log (exponential) phase:** cells divide rapidly at a constant rate.
- **Stationary phase:** nutrients decrease and waste increases, so birth and death rates become equal.
- **Decline (death) phase:** cells die faster than they are produced.

3. Transformation and Transduction

Both are methods of horizontal gene transfer and genetic recombination in bacteria, along with conjugation.

Feature	Transformation	Transduction
How DNA is transferred	Uptake of free DNA from the environment by a living bacterium, usually released from a dead bacterium, which changes its genetic characters	Transfer of bacterial DNA from one bacterium to another through a bacteriophage (virus)
Vector	No vector is involved	A virus acts as the carrier

Pili are used for conjugation, as well as for attachment.

Importance of Bacteria

LESSON 34

1. Economic Importance of Bacteria

Bacteria are economically very important. On the whole, their beneficial uses in industry, agriculture and medicine are enormous. Some bacteria cause diseases, food spoilage and damage to crops, which is their harmful side.

- **Food and industry:** bacteria are used in making curd, cheese, vinegar and alcohol, and in leather tanning and retting of fibres.
- **Medicine:** bacteria are used in making antibiotics such as streptomycin.
- **Agriculture:** nitrogen-fixing bacteria increase soil fertility.
- **Environment:** bacteria are used in sewage treatment and biogas production.

2. Bacterial Diseases in Man

Important bacterial diseases of man include cholera, typhoid, tuberculosis, pneumonia and tetanus. Cleanliness, vaccination and early treatment control these diseases.

- **Cholera:** causes severe watery diarrhoea and dehydration and spreads through contaminated water; it is treated by oral rehydration and antibiotics and prevented by clean water and hygiene.
- **Typhoid:** causes prolonged fever and abdominal pain and is prevented by vaccination and safe food.
- **Tuberculosis:** affects the lungs with chronic cough, fever and weight loss, and is treated with a full course of antibiotics.

3. Bacterial Diseases in Plants

Bacteria cause several plant diseases, such as citrus canker, bacterial blight of cotton and rice, and crown gall. Signs and symptoms include water-soaked spots, cankers, wilting, yellowing, rotting and abnormal growths on leaves, fruits and stems. Infected plants show reduced growth and yield. Treatment includes spraying bactericides and antibiotics and removing and burning infected plants. Prevention is by using disease-free seeds, crop rotation, resistant varieties and disinfection of tools, so the disease does not spread in the field.

Control of Harmful Bacteria

LESSON 35

1. How Cholera and Typhoid Spread

Cholera spreads through water and food contaminated with faeces of an infected person containing Vibrio cholerae, especially during floods and poor sanitation. Houseflies can also transfer the germs to food. Typhoid spreads in a similar faecal-oral way through contaminated water, milk and food, and healthy carriers of Salmonella typhi can spread it unknowingly. Both diseases become epidemics where drinking water is unsafe and sewage disposal is poor. Boiling water and hygienic food handling prevent their spread.

2. Chemical Methods to Control Bacteria

- **Disinfectants and antiseptics:** such as phenol, halogens (chlorine and iodine), heavy metals, soaps and detergents, aldehydes, ethylene oxide and oxidizing agents.
- **Food preservatives:** used as a chemical method of control.
- **Antibiotics:** also used to control bacteria.

3. Physical Methods to Control Bacteria

Physical methods include boiling, dry-heat oven, incineration, autoclaving, pasteurization, refrigeration, freezing, high-pressure processing, desiccation, lyophilization, ionizing and non-ionizing radiation and membrane filtration.

- **Autoclaving:** sterilization by autoclave at 121°C for 15-40 minutes at 15 psi destroys even bacterial spores.

The choice of method depends on the material to be treated.

Protoctists and Fungi

UNIT 7

Protoctists — The Evolutionary Relationships

LESSON 36

1. All Protoctists Are Eukaryotes

All protoctists are **eukaryotes** with a true nucleus, though some are autotrophic and others are heterotrophic.

2. An Artificial Taxon

Protoctista is considered an **artificial taxon** because it is a polyphyletic group made of organisms that do not share a single common ancestor. It includes plant-like algae, animal-like protozoa and fungus-like slime molds, which are very different from each other. They are grouped together mainly because they do not fit into the plant, animal or fungus kingdoms. Biologists therefore regard this kingdom as an unnatural, convenience grouping.

3. A Link Between Prokaryotes and Eukaryotes

Protoctista are regarded as a link because they are the earliest eukaryotes and resemble the ancestors of modern plants, animals and fungi. They show characteristics connecting simple prokaryotic organization with the advanced eukaryotic kingdoms.

- **Plant-like protoctists:** resemble early plants.
- **Protozoa:** resemble early animals.
- **Slime molds:** resemble early fungi.

The three main eukaryotic kingdoms probably arose from protoctist ancestors. Hence protoctists form an evolutionary bridge between prokaryotes and modern eukaryotes.

Major Groups of Protoctists

LESSON 37

1. General Characteristics and Diversity

Protoctista are eukaryotic organisms that do not belong to the plant, animal or fungal kingdoms. They may be unicellular, colonial or simple multicellular, and show autotrophic, heterotrophic, parasitic or absorptive nutrition. They mostly live in water or moist habitats. Because of this diversity, the kingdom is considered polyphyletic.

- **Plant-like protoctists:** the algae, such as Chlorella and Ulva.
- **Animal-like protoctists:** the protozoa, such as Amoeba and Plasmodium.
- **Fungus-like protoctists:** slime molds and water molds, like Phytophthora.

All members of green algae have the pigment combination **chlorophylls a and b**, the same pigment combination as land plants. The amoeboid feeding stage of a slime mold is the multinucleate mass called **plasmodium**. Trypanosoma and Leishmania are disease-causing protoctists (flagellate protozoa) that cause sleeping sickness and kala-azar.

2. Classes of Protozoa — Based on Locomotion

Protozoa are grouped mainly on the basis of their mode of locomotion, e.g. pseudopodia, flagella, cilia or non-motile, and each class is named after its organ of movement.

Class	Movement	Examples
Rhizopoda	Temporary pseudopodia	Amoeba
Flagellata	Whip-like flagella	Euglena, Trypanosoma
Ciliata	Numerous small cilia	Paramecium
Sporozoa	No locomotory organelles in the adult stage; mostly parasites	Plasmodium

3. Ulva — Sea Lettuce

Ulva is commonly called **sea lettuce** because of its green leaf-like thallus found in sea water. It has a flat, green, leaf-like thallus only two cells thick, attached to rocks by a holdfast. Its cells contain a cup-shaped chloroplast and it photosynthesizes like plants.

4. Isomorphic Alternation of Generations in Ulva

Ulva shows **isomorphic alternation of generations**, in which the sporophyte and gametophyte look alike — the two generations are morphologically identical in shape and size. The diploid **sporophyte**, the plant that reproduces asexually, produces haploid zoospores by meiosis, which grow into identical haploid gametophytes. The haploid **gametophyte**, the plant that reproduces sexually, produces gametes that fuse to form a zygote. The zygote grows into a new sporophyte, completing the cycle. The names are based on the type of reproductive cells each generation produces.

General Characteristics of Fungi

LESSON 38

1. Mycology

The study of fungi is called **mycology**.

2. Unusual Nuclear Behaviour in Fungi

Fungi show unusual nuclear behaviour. Their nuclei are haploid for most of the life cycle, and the diploid phase is only the brief zygote stage. During mitosis the nuclear membrane does not break down and the spindle forms inside the nucleus, which is called nuclear (closed) mitosis. In many fungi, two haploid nuclei from different parents coexist in the same cell for a long time — the **dikaryotic condition** — before fusing. Meiosis follows karyogamy immediately during sexual reproduction.

3. Features That Let Fungi Survive Almost Everywhere

Fungi survive in all environments where life is possible because of several features.

- **Spores:** they produce enormous numbers of lightweight spores that spread by air and resist unfavourable conditions, and can remain dormant as spores when conditions are harsh.
- **Hyphae:** hyphae can grow rapidly over and into food sources and absorb nutrients after external digestion.
- **Modes of life:** they can live as saprophytes, parasites or symbionts on a wide range of substrates.
- **Tolerance:** they tolerate a wide range of temperature, pH and moisture.

Diversity Among Fungi

LESSON 39

1. Four Classes of Fungi — Diagnostic Features

Fungi are diverse, and the four classes are distinguished by their hyphae and the way they form spores in sexual reproduction.

Class	Diagnostic Features	Examples
Zygomycota	Non-septate hyphae; form a thick-walled zygospore in sexual reproduction	Rhizopus, Mucor
Ascomycota	Form spores called ascospores inside a sac-like ascus; called sac fungi	Yeast, Aspergillus
Basidiomycota	Form basidiospores on a club-shaped basidium; septate hyphae with a dikaryotic stage	Mushrooms, Agaricus
Deuteromycota	The imperfect fungi, in which sexual reproduction has not been observed; reproduce only asexually by conidia	Penicillium in some classifications

Basidiomycota are the most important cellulose (and wood) degraders in the ecosystem.

In short, Zygomycota are recognised by the thick-walled zygospore, Ascomycota by ascospores formed inside a sac-like ascus, Basidiomycota by basidiospores formed on a club-shaped basidium, and Deuteromycota by the absence of an observed sexual stage, reproducing only asexually by conidia.

Importance of Fungi

LESSON 40

1. Economic Importance of Fungi — Benefits

Fungi are of great economic importance, and their benefits in medicine and industry are immense.

- **Baking and brewing:** yeasts are used in baking and brewing.
- **Medicine:** fungi produce antibiotics like penicillin.
- **Food:** edible mushrooms are a source of food.
- **Decomposition:** fungi are the main decomposers, recycling nutrients in nature; Basidiomycota are the

most important cellulose (and wood) degraders in the ecosystem.

- **Plant growth:** mycorrhizal fungi improve plant growth.
- **Other uses:** some fungi are used in biological control of pests and in biotechnology and genetic research.

2. Economic Importance of Fungi — Harmful Effects

- **Crop diseases:** fungi cause many diseases of crops, e.g. late blight, rusts and smuts.
- **Diseases of humans and animals:** fungi cause diseases of humans and animals.
- **Spoilage and damage:** fungi spoil stored food and damage wood and leather.

Taken together, fungi help humans through food, drink, medicine, nutrient recycling, plant growth, pest control and research, while also harming crops, humans, animals, stored food, wood and leather.

Diversity Among Plants

UNIT 8

Evolutionary Relations in Plants

LESSON 41

1. What Are Plants?

Plants are multicellular photosynthetic eukaryotes with cellulose cell walls, chlorophyll a and b, and alternation of generations.

2. General Characteristics of Plants

- **Cell structure:** Multicellular eukaryotes with cell walls of cellulose and chloroplasts containing chlorophylls a and b.
- **Nutrition:** Mostly autotrophic; they prepare food by photosynthesis and store it as starch.
- **Life cycle:** Shows alternation of a haploid gametophyte and a diploid sporophyte generation.
- **Movement and range:** Most plants are fixed in one place and show limited movement, ranging from tiny mosses to giant trees.

3. Evolutionary Relation in Plants

Land plants arose from green algal ancestors. Bryophytes were the earliest land plants without vascular tissue, followed by vascular seedless plants like ferns. Gymnosperms then evolved naked seeds, and finally angiosperms evolved flowers, enclosed seeds and double fertilization, becoming the most successful group.

The rapid diversification of land (vascular) plants occurred in the Devonian period.

4. Alternation of Generations

The life cycle alternates between a haploid gametophyte and a diploid sporophyte. The sporophyte produces haploid spores by meiosis, which grow into the gametophyte. The gametophyte produces gametes, which fuse to form a diploid zygote that grows into a new sporophyte. This alternation is found in all land plants, though the dominant generation differs.

5. How Plants Coped on Land

On invading land from the sea, plants faced drying, temperature changes and lack of support. They developed a waxy cuticle and stomata to control water loss, and spores with sporopollenin walls — sporopollenin makes spore and pollen walls tough, flexible and resistant to biodegradation. Vascular tissues and lignin gave conduction and support, roots anchored them, and protected embryos and, later, seeds and pollen allowed reproduction without free water.

6. Heterogamy and Oogamy

Heterogamy is important because the production of two different types of gametes allows specialization. The small motile male gamete can travel to the female gamete, while the large female gamete stores food for the early embryo. This division of labour increases the chances of successful fertilization and survival of the offspring. Heterogamy reached its most successful form, oogamy — a large non-motile egg and small motile sperm — in land plants, the most successful kind of reproduction, and led to seed formation.

Non-Vascular Plants — Bryophytes

LESSON 42

1. Bryophytes — The Non-Vascular Plants

Hepaticae (liverworts), Anthocerotae (hornworts) and Mosses are the classes of Bryophytes. Bryophytes were the earliest land plants without vascular tissue.

2. Why Bryophytes Are Called Non-Vascular Plants

Bryophytes are called non-vascular plants because they lack true vascular tissues, xylem and phloem, for conduction of water and food. They also lack true roots, stems and leaves, having instead rhizoids and simple leaf-like structures. Water moves in them slowly by diffusion from cell to cell. For this reason bryophytes remain small and live in moist places.

3. Life Cycle of Moss

In moss, the dominant free-living generation is the haploid gametophyte, which grows from a spore through a filamentous protonema. The mature gametophyte bears antheridia producing male gametes and archegonia producing eggs; fertilization needs water and forms a diploid zygote. The zygote grows into a sporophyte (foot, seta and capsule) that remains attached to and dependent on the gametophyte. The capsule produces haploid spores by meiosis, which are dispersed by wind and germinate to repeat the cycle.

4. Bryophytes Compared with Vascular Plants

Feature	Bryophytes	Vascular Plants
Vascular tissue	No true xylem and phloem	Xylem and phloem present for conduction
Body parts	No true roots, stems and leaves; rhizoids present	True roots, stems and leaves
Dominant generation	Gametophyte (in moss)	Sporophyte
Size and habitat	Small; live in moist places	Can grow tall; inhabit diverse and dry habitats

Seedless Vascular Plants

LESSON 43

1. Seedless Vascular Plants

Vascular seedless plants like ferns followed the bryophytes in plant evolution. In living vascular plants the sporophyte is the dominant, free-living generation.

Psilophyta (e.g. Psilotum) are the most primitive living vascular plants, lacking true roots and leaves.

2. The Fern Gametophyte

The free-living heart-shaped gametophyte of a fern is called the prothallus.

3. Why Vascular Plants Are More Successful Land Plants

Vascular plants are considered more successful land plants because their xylem and phloem conduct water, minerals and food efficiently over long distances. Lignified xylem also gives mechanical strength, allowing them to grow tall and compete for light. They have true roots, stems and leaves, a dominant sporophyte, and in seed plants, pollen and seeds that free reproduction from water. These adaptations let them inhabit diverse and dry habitats and dominate the land flora.

4. How Leaves Evolved

Leaves evolved in two main ways during plant evolution:

- **Microphylls:** As in Lycophyta, developed as small outgrowths (enations) of the stem that later received a single vascular strand.
- **Megaphylls:** Found in ferns and seed plants, evolved from flattened and webbed branching systems: side branches became flat, arranged in one plane, and the spaces between them filled with photosynthetic tissue. Megaphylls have a branched vein system and a large surface for photosynthesis.

This evolution greatly increased the photosynthetic efficiency of land plants.

Seed Plants — Gymnosperms and Angiosperms

LESSON 44

1. How Seeds Evolved

Seeds evolved from the heterospory of earlier plants, in which two types of spores, microspores and megaspores, were produced. The megaspore was retained inside the megasporangium (ovule) on the parent plant instead of being shed. A protective integument developed around the ovule, and pollen delivered the male gamete without needing water. After fertilization the ovule became the seed, containing an embryo with stored food. This gave plants a great advantage for survival and dispersal on land.

2. Gymnosperms — Naked Seeds

Gymnosperms have naked seeds because their ovules and seeds are not enclosed in an ovary; they lie exposed on cone scales. The word gymnosperm itself means 'naked seed'. Gymnosperms evolved naked seeds before angiosperms evolved flowers and enclosed seeds.

Resins, turpentine and pine oil are obtained from pines (Pinus).

3. The Flower — Reproductive Part of the Plant

The flower is called the reproductive part of the plant because it bears the male and female reproductive organs. The stamen (anther and filament) produces pollen grains containing male gametes, and the carpel (stigma, style and ovary) contains ovules with the egg. Pollination and fertilization occur through the flower, after which the ovary forms fruit and ovules form seeds.

In banana, the inflorescence/flowers are covered by one or many large bracts called spathes.

4. Double Fertilization in Angiosperms

Double fertilization is the characteristic feature of angiosperms (flowering plants). Fertilization in angiosperms is called double fertilization because two fusions occur inside the embryo sac. One male gamete fuses with the egg to form the diploid zygote (syngamy). The second male gamete fuses with the two polar nuclei to form a triploid primary endosperm nucleus, which develops into endosperm, the food tissue for the embryo. Since two fertilization events take place together, it is called double fertilization.

5. Gymnosperms Compared with Angiosperms

Feature	Gymnosperms	Angiosperms
Seeds	Naked; exposed on cone scales, not enclosed in an ovary	Enclosed inside the carpel's ovary, which becomes the fruit wall
Reproductive structure	Cones	Flowers
Fertilization	Single fertilization	Double fertilization

Diversity Among Animals

UNIT 9

Characteristics of Animals and Criteria of Classification

LESSON 45

1. Characteristics of Animals

Animals are multicellular, heterotrophic eukaryotes without cell walls. Among the terms listed, the (Hox) genes that control animal body patterning are properly associated only with animals; cell wall and autotrophy are absent in animals, and sexual reproduction also occurs in plants and fungi.

2. How Animals Are Classified

Animals are classified into about 33 phyla on the basis of their evolutionary relationships and body organization. Only nine major phyla, from Porifera to Chordata, are usually studied in depth, while the rest are minor phyla. Comparative morphology and embryology provide the main clues.

3. Basic Factors of Classification

- **Level of organization:** cellular, tissue, organ or organ-system level.
- **Number of germ layers:** diploblastic or triploblastic.
- **Symmetry:** asymmetrical, radial or bilateral.
- **Coelom:** animals may be acoelomate, pseudocoelomate or true coelomate.
- **Fate of the blastopore:** divides triploblastic coelomates into protostomes and deuterostomes.
- **Notochord:** presence of a notochord separates chordates from invertebrates.

An adult animal that possesses bilateral symmetry is also triploblastic, with three germ layers, though it may not be coelomate, protostome or deuterostome. Bilateria encompasses all the others, as coelomates, protostomes and triploblastic animals are all subgroups of bilaterally symmetrical animals.

4. Diploblastic and Triploblastic Animals

Diploblastic	Triploblastic
Develop from only two germ layers, ectoderm and endoderm, with a jelly-like mesoglea between them, such as cnidarians.	Develop from three germ layers, ectoderm, mesoderm and endoderm. The mesoderm allows formation of complex organs, muscles and, in many, a coelom, so they show a much higher level of organization.

5. Protostomes and Deuterostomes

Protostomes	Deuterostomes
The blastopore, the first opening of the embryo, develops into the mouth; the word protostome means 'mouth first' and the anus is formed later at a separate place. They generally show spiral and determinate cleavage, and the mouth develops from or near the blastopore. Molluscs, annelids and arthropods belong to this group.	The blastopore becomes the anus and the mouth is formed secondarily.

Porifera and Cnidaria

LESSON 46

1. Phylum Porifera — Sponges

Sponges are considered the simplest animals, in phylum Porifera. They are placed in the animal kingdom because they are multicellular, heterotrophic eukaryotes without cell walls, like other animals. They ingest food particles, have animal-like cells such as choanocytes, and reproduce sexually producing a free-swimming larva. Although they have only a cellular level of organization and lack true tissues, their nutrition, structure and development clearly show animal affinities.

2. Spicules and Water Movement

- **Sponges:** associated with spicules, their structural material.
- **Spicule formation:** amoebocytes (scleroblasts) manufacture the spicules that form the skeleton / structural material of sponges.

- **Water movement:** water enters the sponge through ostia, passes into the spongocoel and leaves through the osculum, i.e. Ostia → Spongocoel → Osculum.

3. Phylum Cnidaria

Cnidaria are associated with nematocysts. Cnidarians are diploblastic animals, developing from only two germ layers, ectoderm and endoderm, with a jelly-like mesoglea between them.

- **Cnidaria:** nematocysts.
- **Germ layers:** two — ectoderm and endoderm.
- **Mesoglea:** jelly-like layer between the two germ layers in diploblastic animals.

Platyhelminthes and Aschelminthes

LESSON 47

1. Phylum Platyhelminthes — Flatworms

Flatworms (phylum Platyhelminthes) are named so because their bodies are dorsoventrally flattened like a ribbon or leaf. This flat shape keeps every cell close to the surface, so gases and food can diffuse easily, as they have no circulatory system. They are triploblastic, bilaterally symmetrical and acoelomate animals.

- **Examples:** Planaria, tapeworm and liver fluke.
- **Body form:** dorsoventrally flattened like a ribbon or leaf.
- **Organisation:** triploblastic, bilaterally symmetrical and acoelomate.

2. Aschelminthes — Nematodes

Nematodes are distinguished from annelids by the type of body cavity, the number of muscle layers in the body wall, and the presence of segmentation.

Feature	Nematodes	Annelids
Type of body cavity	Pseudocoelom	True coelom
Number of muscle layers in the body wall	Differs from annelids	Differs from nematodes
Segmentation	Absent in nematodes	Present in annelids
Number of embryonic tissue layers	Not used to distinguish nematodes from annelids in the given criteria — the distinguishing points are I, II and III above	

The flatworms are acoelomate, while nematodes have a pseudocoelom, showing a different type of body cavity from both flatworms and annelids.

Annelida and Mollusca

LESSON 48

1. Phylum Annelida — Segmented Worms

Segmented worms (annelids) move by setae / parapodia, not pseudopodia; pseudopodia belong to Amoeba, so the pair segmented worm → pseudopodia is incorrect.

2. Evolutionary Adaptations of Annelida

- **Metameric segmentation:** the body is metamerically segmented, with repeating units that allow specialization of different body regions.
- **True coelom:** acts as a hydrostatic skeleton and permits efficient burrowing movement.
- **Circulatory system:** closed circulatory system.
- **Digestive tract:** complete digestive tract.
- **Nervous and excretory systems:** well-developed nervous and excretory systems, with nephridia in each segment.
- **Locomotion:** setae and, in polychaetes, parapodia help in locomotion.

These advances make annelids successful in marine, freshwater and moist land habitats, and show important evolutionary advances over simpler worms. Annelids belong to the protostomes.

3. Phylum Mollusca

Molluscs are soft-bodied, unsegmented, triploblastic coelomate animals, usually protected by a shell of calcium carbonate secreted by the mantle. Molluscs also belong to the protostomes.

- **Head:** part of the body division.
- **Muscular ventral foot:** used for locomotion.
- **Visceral mass:** containing the organs.
- **Radula:** a unique rasping feeding organ in the mouth in most molluscs.
- **Respiration:** by gills or mantle cavity.
- **Circulatory system:** mostly open circulatory system.
- **Examples:** snails, mussels, octopus and squid.

Arthropoda and Echinodermata

LESSON 49

1. Phylum Arthropoda

Arthropods are triploblastic, bilaterally symmetrical, coelomate animals with a segmented body divided into head, thorax and abdomen (or cephalothorax and abdomen). They belong to the protostomes.

- **Jointed appendages:** arthropods have jointed legs / appendages, from which the phylum gets its name.
- **Exoskeleton:** a hard exoskeleton of chitin that is moulted during growth.
- **Digestive system:** complete digestive system.
- **Circulatory system:** open circulatory system.
- **Respiration:** by gills, tracheae or book lungs.
- **Size of phylum:** the largest phylum, including insects, spiders, crustaceans and myriapods, adapted to every habitat.

2. Phylum Echinodermata

Star fish, brittle star and sea urchin are echinoderms.

3. Why Echinoderms Are Placed in Bilateria

Echinoderms are placed in Bilateria because their larvae are bilaterally symmetrical, showing their true evolutionary relationship. The radial symmetry of adults is a secondary adaptation to their slow-moving or sessile bottom-dwelling life. Their development is deuterostome-like, similar to chordates, which also supports this placement. Adult radial symmetry therefore does not reflect their ancestry.

- **Examples:** star fish, brittle star and sea urchin.
- **Larvae:** bilaterally symmetrical.
- **Adults:** radially symmetrical as a secondary adaptation.
- **Development:** deuterostome-like, similar to chordates.

Chordata — Fishes and Mammals

LESSON 50

1. Phylum Chordata

Both invertebrate chordates and vertebrates show chordate characters at some stage of life. Lamprey is a vertebrate (jawless fish), not an echinoderm. Dry scaly skin, ectothermic, two pairs of legs, lungs, internal fertilization, eggs with amniotic shell, and a three-chambered heart are all properties of class Reptilia.

2. Invertebrate Chordates and Vertebrates

Invertebrate Chordates	Vertebrates
Possess a notochord but no vertebral column or backbone; small and mostly marine. Examples are Amphioxus, tunicates and hagfish-like forms in the subphyla Urochordata and Cephalochordata.	Have the notochord replaced by a vertebral column made of vertebrae, a well-developed skull and brain, and an internal skeleton. They include fishes, amphibians, reptiles, birds and mammals.

3. Cartilaginous Fishes and Bony Fishes

Cartilaginous Fishes	Bony Fishes
Skeleton made of cartilage, placoid scales, a heterocercal tail and no swim bladder or operculum; e.g. sharks, rays. Most are marine.	Skeleton mainly of bone, cycloid or ctenoid scales, a homocercal tail and a gill cover (operculum); e.g. rohu, trout. They live in both marine and fresh water.

Fertilization and other habits also differ between the two classes of fishes.

4. Mammals

Mammals are warm-blooded vertebrates that feed their young on milk from mammary glands, have hair or fur on the body, and possess a diaphragm and a four-chambered heart.

- **Subclass Prototheria (monotremes):** lay eggs, have mammary glands without nipples and a cloaca; e.g. echidna and duck-billed platypus.
- **Subclass Metatheria (marsupials):** give birth to tiny immature young that complete development in a mother's pouch (marsupium); e.g. kangaroo and koala.
- **Subclass Eutheria:** placental mammals whose young develop fully inside the uterus nourished through a placenta; e.g. human, whale, bat and lion.

Forms and Functions in Plants

UNIT 10

Mineral Nutrition in Plants

LESSON 51

1. Where Plants Get Their Nutrients

Plants primarily obtain their essential mineral nutrients from the soil through their roots. Mineral nutrients are necessary because plants use them to build essential compounds and to carry out metabolism. Although needed in different amounts, each essential element has a specific role that no other element can fully replace. Without these minerals, plants show deficiency symptoms, poor growth and low yield.

2. Work Done by Some Nutrients

- **Nitrogen:** needed for proteins and chlorophyll, and also for amino acids and nucleic acids.
- **Phosphorus:** needed for ATP, nucleic acids and membranes.
- **Potassium:** needed for enzyme activation and stomatal working.

3. Macro and Micro Nutrients

Essential inorganic nutrients are divided into macronutrients, needed in large amounts, and micronutrients, needed in trace amounts. Iron is a micronutrient, needed in trace amounts, while potassium, phosphorus and sulphur are macronutrients.

Macronutrients (large amounts)	Micronutrients (trace amounts)
Nitrogen, phosphorus, potassium, sulphur, calcium, magnesium (and sodium, chlorine in the book's list)	Iron, manganese, boron, zinc, copper, molybdenum and cobalt

4. General Functions of Inorganic Nutrients

In general, inorganic nutrients form structural parts of proteins, chlorophyll, nucleic acids and membranes. They also act as enzyme activators and cofactors, maintain osmotic balance and pH, and take part in energy transfer, e.g. phosphorus in ATP. Deficiency of any essential nutrient disturbs growth.

5. Why Nitrogen Is Counted Among Mineral Nutrients

Nitrogen is included among mineral nutrients although it is not itself a mineral obtained as a metal, because plants absorb it from the soil in mineral (inorganic) forms, as nitrates and ammonium ions. It is an essential macronutrient required in large amounts for amino acids, proteins, nucleic acids and chlorophyll. Atmospheric N_2 cannot be used directly by most plants and must first be fixed into these mineral forms. For this reason nitrogen is listed with the mineral nutrients.

6. A Deficiency Sign to Remember

Deficiency of phosphate causes leaves to turn dull or bluish green (often with purplish tint) and growth is stunted.

Carnivorous Plants

LESSON 52

1. Plants That Catch Insects

Carnivorous plants typically live in nitrogen-deficient soils. They grow in soils that are deficient in nitrogen and other minerals, so they cannot get enough nitrogen from their roots.

2. Why They Use Insects as Food

Carnivorous plants trap and digest insects to obtain nitrogen and other nutrients from insect proteins. Insects only supplement their mineral nutrition. They still make their own food by photosynthesis; insects are used because the soil cannot supply enough nitrogen through the roots.

3. Leaves Modified into Traps

- **Trap leaves:** their leaves are modified into traps.
- **Forms of trap:** the traps are like pitchers and snap traps.
- **Digestion:** the traps work with digestive enzymes, and the plant digests the trapped insects.

- **Nutrients gained:** nitrogen and other nutrients are obtained from insect proteins.

4. The Whole Picture

Feature	Position in Carnivorous Plants
Soil	Deficient in nitrogen and other minerals
Roots	Cannot get enough nitrogen from the soil
Photosynthesis	Still carried out — the plant makes its own food
Insects	Trapped and digested to supplement mineral nutrition

In short, carnivorous plants live in mineral-deficient soil, especially nitrogen-deficient soil, and meet the shortage by digesting insects, while photosynthesis continues as their way of making food.

Transport in Plants — Ascent of Sap and Translocation of Sugars

LESSON 53

1. Two Transport Systems

Water moves up through the xylem in the ascent of sap, and sugars move through the phloem from a source to a sink. Active transport of sucrose into sieve elements at the source is called phloem loading.

2. Cohesion-Tension Theory — Ascent of Sap in Xylem

The cohesion-tension theory explains the ascent of sap in xylem. Transpiration from leaves creates a tension (negative pressure) at the top, which is the source of the pulling force. Water molecules cohere to each other by hydrogen bonds, forming a continuous unbroken column in the xylem, and also adhere to xylem walls. The transpiration pull is transmitted down this cohesive column, drawing water from roots to the tallest leaves. Thus cohesion keeps the column intact while tension pulls it upward.

- **Cohesion:** supplied by hydrogen bonds between water molecules; it keeps the column intact.
- **Tension:** its source is transpiration from leaves; it pulls the column upward.

3. Pressure Flow Theory — Movement of Sugar

According to the pressure flow theory, sucrose is actively loaded into sieve tubes at the source (leaf), lowering their water potential. Water enters by osmosis from the xylem, creating high turgor pressure that pushes the sap towards the sink (root, fruit or growing part). At the sink, sucrose is unloaded and used or stored, water leaves the sieve tubes, and pressure falls. This pressure difference between source and sink drives the mass flow of sugars through the phloem.

Step	What Happens
Phloem loading	Sucrose is actively loaded into sieve tubes at the source; water potential falls
Water entry	Water enters by osmosis from the xylem; high turgor pressure is created
Mass flow	Sap is pushed towards the sink
Unloading	Sucrose is unloaded and used or stored; water leaves and pressure falls

Gaseous Exchange in Plants and Osmotic Adjustment

LESSON 54

1. Exchange of Gases in Plants

Gaseous exchange in plants occurs mainly through stomata in leaves and lenticels in woody stems, and through the general surface of roots. During the day, CO_2 enters through open stomata for photosynthesis and oxygen produced is released. For respiration, oxygen diffuses in and CO_2 diffuses out, day and night, according to concentration gradients. Guard cells regulate stomatal opening to balance CO_2 uptake with water loss by transpiration. Intercellular spaces inside the leaf connect the stomata to all photosynthetic cells, making exchange efficient.

2. Water Potential and Osmotic Adjustment

Water potential describes the capacity/tendency of a living system to lose (or gain) water. In osmotic adjustment, plant cells accumulate solutes such as salts, sugars and amino acids in their vacuoles and cytoplasm. This lowers the cell's osmotic (water) potential, so the plant can draw and retain water even when soil water is scarce or salty. It helps maintain turgor, keeps stomata functioning and allows growth to continue during drought. Plants showing good osmotic adjustment are therefore more tolerant of water stress and salinity.

3. Plants Categorised by Water Relations

Category	Features
Hydrophytes	Live in water, have little need for osmotic adjustment, thin cuticle and large air spaces
Mesophytes	Live in moderate water conditions and show ordinary regulation of stomata and turgor
Xerophytes	Plants of deserts; adjust osmotically by accumulating solutes, reducing leaves and storing water to survive drought
Halophytes	Plants growing in salt marshes close to the sea; accumulate salts in vacuoles and may excrete excess salt, maintaining water uptake from saline soil

4. Why Desert Plants Reduce Leaf Size

Desert plants reduce their leaf size to decrease the surface area available for transpiration, so that water loss is minimized in hot and dry conditions. Small leaves, or leaves changed into spines as in cactus, also reduce heating of the plant. Photosynthesis is then often carried out by the green stem. This adaptation, together with thick cuticle and sunken stomata, helps desert plants conserve water.

Growth Regulators — Plant Hormones

LESSON 55

1. What Growth Regulators Are

Growth regulators (plant hormones) are organic substances produced in very small amounts that control growth and development. Together these five regulators coordinate the plant's life activities.

2. The Principal Natural Auxin

IAA (indole acetic acid) is the principal natural auxin of higher plants, synthesized at the apices of stem and root. The production of lateral appendages (leaves and branches) depends on the activity of the apical meristem.

3. The Five Growth Regulators

Regulator	Functions
Auxins (e.g. IAA)	Promote cell elongation, root initiation and apical dominance
Gibberellins	Promote stem elongation, seed germination and flowering
Cytokinins	Promote cell division and delay leaf ageing, working with auxins in tissue growth
Abscisic acid	Inhibits growth, promotes dormancy and closes stomata during water stress
Ethene (ethylene)	Promotes fruit ripening, leaf fall and senescence

4. How the Five Work Together

- **Auxins:** cell elongation, root initiation and apical dominance.
- **Gibberellins:** stem elongation, seed germination and flowering.
- **Cytokinins:** cell division, delayed leaf ageing, and tissue growth with auxins.
- **Abscisic acid:** growth inhibition, dormancy, and stomatal closing during water stress.
- **Ethene:** fruit ripening, leaf fall and senescence.

Some regulators mainly promote growth processes such as elongation, division and germination, while abscisic acid inhibits growth and promotes dormancy, and ethene promotes ripening, leaf fall and senescence.

Photoperiodism, Vernalization and Annual Rings

LESSON 56

1. Photoperiodism and the Phytochrome Pigment

Phytochrome is the light-sensitive pigment that enables plants to measure day and night length, which is the basis of photoperiodism. It exists in two interconvertible forms, Pr and Pfr, which change with red and far-red light. The amount of active Pfr at the end of the night informs the plant whether the critical night length has passed. Accordingly, short-day and long-day plants initiate flowering at the proper season.

2. Vernalization — Flowering Promoted by Cold

Promotion of flowering by cold treatment is called vernalization.

3. Why Cold Treatment Is Needed for Germination in Some Plants

In some plants, seeds remain dormant even when water and oxygen are available, because growth inhibitors are present or the embryo is not fully ready. Exposure to a period of low temperature (cold treatment/stratification) breaks this dormancy by changing the balance of hormones, decreasing abscisic acid and increasing gibberellins. This ensures that seeds germinate in spring, not in the unfavourable winter. Hence cold treatment is necessary for germination of such seeds.

4. How Annual Rings Are Formed

Annual rings are formed by the seasonal activity of the vascular cambium during secondary growth. In spring, cambium produces large, thin-walled xylem cells (early wood), while in late summer and autumn it produces smaller, thick-walled cells (late wood). The sharp contrast between late wood of one year and early wood of the next appears as a ring. Counting these rings gives the approximate age of the tree.

Wood	Season Formed	Cells
Early wood	Spring	Large, thin-walled xylem cells
Late wood	Late summer and autumn	Smaller, thick-walled cells

Holozoic Nutrition

UNIT 11

Stages of Holozoic Nutrition and Types of Feeders

LESSON 57

1. Stages of Holozoic Nutrition

Holozoic nutrition consists of five stages that occur in sequence in animals like man. Removal of undigested food is egestion, a separate elimination step; nutrition proper consists of ingestion, digestion, absorption and assimilation together with egestion as the final stage.

- **Ingestion:** the taking in of food through the mouth.
- **Digestion:** the mechanical and chemical breakdown of complex food into simple soluble molecules.
- **Absorption:** the uptake of these molecules through the intestinal lining into blood.
- **Assimilation:** the use of absorbed food for energy, growth and repair.
- **Egestion:** the removal of undigested food from the body.

Enzymes play the most important role in digestion, as they chemically break down complex food into simple soluble molecules during digestion.

2. Teeth and Taking in Food

The thecodont condition means teeth embedded in sockets in the jaw bone, as in mammals and crocodiles. A child has 20 milk (deciduous) teeth, which are later replaced by permanent teeth. Incisors are sharp, chisel-shaped teeth meant for cutting and biting food. The pharynx is a common passage for food and air, so the part of the gut also associated with the respiratory system is the pharynx.

3. Types of Feeders — By Size of Food

On the basis of the size of food they take, animals are categorized into three groups. This grouping reflects different methods of ingestion in holozoic animals.

- **Microphagous:** animals feed on tiny particles, e.g. Amoeba, Paramecium and many filter-feeding invertebrates.
- **Macrophagous:** animals take in large pieces of food or whole organisms, e.g. Hydra, frog and man.
- **Liquid feeders:** animals suck fluids, e.g. mosquitoes and leeches suck blood, butterflies suck nectar and spiders suck liquefied prey.

Digestion in Amoeba and Types of Digestive Systems

LESSON 58

1. Digestion in Amoeba

Amoeba shows intracellular digestion. When Amoeba contacts a food particle, it extends pseudopodia around it and engulfs it by phagocytosis, forming a food vacuole. Lysosomes fuse with the vacuole and pour digestive enzymes into it, first making it acidic and then alkaline. The food is digested inside the vacuole and the soluble products are absorbed into the surrounding cytoplasm and assimilated. The undigested residue is removed by egestion when the vacuole reaches the cell surface and opens out.

2. Intracellular and Extracellular Digestion

Planaria shows both extracellular digestion in the gut cavity and intracellular digestion inside food vacuoles of gut cells.

Feature	Intracellular Digestion	Extracellular Digestion
Site	Inside food vacuoles of cells	In the gut cavity
Example	Amoeba; also inside gut cells of Planaria	Gut cavity of Planaria and higher animals

3. Sac-like Gut

A sac-like gut is an incomplete digestive system with only one opening that serves as both mouth and anus. Food enters through the mouth into a sac-like gastrovascular cavity, where extracellular and intracellular digestion take place. Undigested residue is egested through the same opening, so feeding and egestion cannot occur at the same time. Hydra and Planaria are suitable examples of animals with a sac-like gut. It is simpler than the tube-like gut of higher animals, which has separate mouth and anus.

Feature	Sac-like Gut	Tube-like Gut
Openings	Only one opening — both mouth and anus	Separate mouth and anus
Cavity	Sac-like gastrovascular cavity	Tube-like alimentary canal
Feeding and egestion	Cannot occur at the same time	Can proceed through separate openings
Examples	Hydra, Planaria	Higher animals

Digestion in the Stomach

LESSON 59

1. Digestion in the Stomach

In the stomach, food is churned by muscular contractions and mixed with gastric juice to form a semi-liquid chyme. Rugae are the folds of the inner wall of the stomach that allow it to expand and help in churning food. Stomach digestion is mainly of proteins.

Gastric juice contains HCl, which kills germs, provides an acidic medium and converts inactive pepsinogen into pepsin. Pepsin begins protein digestion, breaking proteins into peptones, while in infants rennin curdles milk protein. Partly digested food is gradually released into the duodenum through the pyloric sphincter.

2. How the Stomach is Protected Against Strong HCl

The stomach is protected from its own strong HCl and pepsin by a thick layer of mucus secreted by its inner wall, which forms a physical barrier. The epithelial cells of the stomach lining are tightly joined and are replaced rapidly when damaged. Pepsin is secreted in an inactive form, pepsinogen, and is activated only in the lumen. Bicarbonate in the mucus also neutralizes acid near the wall. Together these protections prevent autodigestion of the stomach.

- **Mucus barrier:** thick mucus layer secreted by the inner wall forms a physical barrier.
- **Tight epithelial lining:** epithelial cells are tightly joined and replaced rapidly when damaged.
- **Inactive enzyme:** pepsin secreted as pepsinogen, activated only in the lumen.
- **Bicarbonate:** bicarbonate in the mucus neutralizes acid near the wall.

Digestion and Absorption in the Small Intestine

LESSON 60

1. Digestion in the Small Intestine

In the small intestine, chyme is mixed with bile, pancreatic juice and intestinal juice in the duodenum and jejunum, and digestion of all food types is completed here. Pancreatic amylase, lipase and proteases and intestinal enzymes convert carbohydrates to glucose, fats to fatty acids and glycerol, and proteins to amino acids.

2. Absorption in the Small Intestine

The products are absorbed through the villi: glucose, amino acids, vitamins and salts enter blood capillaries, while fatty acids and glycerol enter the lacteal (lymph) of the villus. The enormous surface provided by folds, villi and microvilli makes absorption highly efficient.

3. How the Internal Surface Area is Increased

The internal surface area of the small intestine is greatly increased in three ways. These structures multiply the absorptive area many times. A large surface area, thin walls and a rich blood and lymph supply make absorption of digested food very efficient.

- **Circular folds:** the inner wall has circular folds.
- **Villi:** the surface bears millions of finger-like villi.
- **Microvilli:** each epithelial cell of a villus has tiny microvilli forming a brush border.

Accessory Glands and the Large Intestine

LESSON 61

1. Accessory Glands of the Gut

The accessory glands of the human gut are the salivary glands, liver and pancreas, along with glands in the stomach and intestinal walls. Together these glands make complete chemical digestion possible.

- **Salivary glands:** secrete saliva containing amylase that starts starch digestion and moistens food. There are six salivary glands, i.e. three pairs (parotid, submaxillary/submandibular and sublingual), opening into the oral cavity.
- **Liver:** produces bile, which emulsifies fats, and also processes absorbed nutrients carried by the hepatic portal vein.
- **Pancreas:** secretes pancreatic juice with enzymes for proteins, fats and carbohydrates and also insulin.
- **Gastric and intestinal glands:** add their own juices.

2. How Bile Helps in Digestion of Fats

Bile, produced by the liver and stored in the gall bladder, contains bile salts but no enzymes. Bile salts break large fat globules into tiny droplets, a process called emulsification, which greatly increases the surface area of fats. This allows the enzyme lipase to digest fats quickly into fatty acids and glycerol. Bile also neutralizes the acidic chyme from the stomach, providing the alkaline medium needed by intestinal enzymes.

3. Role of the Large Intestine

The large intestine mainly absorbs water and salts from the liquid material received from the small intestine, converting it into semi-solid faeces. It also absorbs some vitamins produced by its bacteria, such as vitamin K and B vitamins. Its mucus lubricates the passage of faeces. Finally, it stores faeces temporarily and removes them by egestion through the anus. It has no role in enzymatic digestion of food.

4. Bacteria in the Large Intestine

Bacteria reach the large intestine with food and water and find there a warm environment, moisture and a continuous supply of undigested material on which to feed. Unlike the stomach, the large intestine has no strong acid to kill them, so they multiply and live as a normal flora. These bacteria are mostly harmless or useful; they produce vitamins and gases and help form faeces. Their presence is therefore a normal feature of the gut.

Eating Disorders — Obesity and Anorexia Nervosa

LESSON 62

1. Obesity

According to WHO, a body mass index (BMI) over 30 kg/m² in an adult is considered obesity.

2. Factors Which May Lead to Obesity

Obesity develops from a combination of factors. Obesity results from inheritance, environment, diet and lack of exercise combined. When energy intake remains greater than energy expenditure for a long time, fat accumulates and obesity results.

- **Genetic inheritance:** can make a person prone to storing fat.
- **Environment:** an environment with abundant energy-rich food and little physical work encourages weight gain.
- **Diet:** eating excess fats and sugars, and overeating, are major causes.
- **Sedentary lifestyle:** lack of exercise is a major cause.
- **Other factors:** hormonal disorders and certain medications may also contribute.

3. Health Risks Involved in Obesity

Obesity increases the risk of many serious diseases. It is linked with type 2 diabetes, high blood pressure, heart and cardiovascular diseases, stroke and certain cancers. Excess weight also strains joints, causing arthritis, and can disturb breathing during sleep. Long-term proper diet and regular physical activity are needed to control it.

4. Anorexia Nervosa

Anorexia nervosa is a psychological eating disorder in which a person has an intense fear of gaining weight and refuses to eat adequately. It is most common in young girls and women. The victim loses appetite and weight falls well below normal; some patients also practise induced vomiting. It can cause severe weakness, hormonal disturbance and even death if untreated. Treatment needs psychological help, medical care and proper nutrition.

Circulation

UNIT 12

Circulation — Need and Types

LESSON 63

1. What Is Circulation?

Circulation is the transport of materials within the body through the heart and blood vessels, and in many animals through a lymphatic system as well. Cells continuously require oxygen and nutrients for metabolism and produce wastes that must be removed.

2. Need in Higher Animals

Higher animals are large and multicellular, so simple diffusion cannot supply all cells quickly enough: cells lie far from the body surface and from the gut, the rate of metabolism is high, and diffusion alone is too slow. Circulation is essential for survival of every cell.

- **Transport:** oxygen, nutrients, hormones and waste products rapidly to and from every cell through blood vessels.
- **Other roles:** distributes hormones, maintains body temperature, keeps pH and fluid balance, and carries antibodies and white blood cells for defence.

Circulation only transports wastes to the excretory organs; actual removal/excretion is performed by the kidneys etc., not by circulation itself.

3. Open and Closed Circulation

In a **closed** system blood is confined to vessels and remains distinct from the interstitial fluid surrounding the cells.

Arthropods have an **open** system: blood is not always confined to vessels. The heart pumps it through short arteries into open spaces (sinuses) of the body cavity, where it directly bathes the internal organs. There is no distinction between blood and interstitial fluid, so the general body fluid is called haemolymph. Blood returns slowly to the heart through small openings called ostia.

4. Single and Double Circuit Circulation

	Single Circuit	Double Circuit
Heart passage	Once in one complete circuit; heart two-chambered, as in fishes	Twice — pulmonary circuit to lungs and systemic circuit to body, as in amphibians to mammals
Efficiency	Less efficient oxygen supply to tissues	Oxygenated and deoxygenated blood largely separate in birds and mammals; more efficient

5. Pulmonary and Systemic Circulation

Pulmonary: deoxygenated blood from the right ventricle through the pulmonary artery to the lungs, where it is oxygenated, returning through the pulmonary veins to the left atrium. It serves gaseous exchange; oxygenated blood in a pulmonary vein drains first into the left atrium.

Systemic: oxygenated blood from the left ventricle through the aorta to all body tissues, returning deoxygenated through the vena cava to the right atrium. It supplies the whole body. Together they form double circulation.

The Human Heart — Structure, Working and ECG

LESSON 64

1. Structure of the Human Heart

A somewhat conical, muscular organ enclosed in the pericardium. Four chambers: two upper thin-walled atria and two lower thick-walled ventricles. Right and left sides are separated by inter-atrial and inter-ventricular septa, so oxygenated and deoxygenated blood do not mix. Tricuspid and bicuspid valves guard the atrioventricular openings; semilunar valves guard the aorta and pulmonary artery. The wall is cardiac muscle (myocardium), covered externally and lined internally. The SA node in the right atrium acts as pacemaker.

2. SA Node — Pacemaker

It initiates each heartbeat, generating electrical impulses at a regular rhythm that spread over the atria, making them contract, and then reach the AV node and ventricles. It sets the normal rate and timing of pumping, about 70-75 times per minute at rest. Damage to the S.A node would disrupt the rate and timing of cardiac muscle contractions.

3. Flow of Blood Through the Heart

Body → vena cava → right atrium → tricuspid valve → right ventricle → semilunar valve → pulmonary artery → lungs → pulmonary veins → left atrium → bicuspid (mitral) valve → left ventricle → semilunar valve → aorta → body.

When ventricles contract, the tricuspid and bicuspid valves close to prevent backflow and blood leaves through the semilunar valves, located where blood leaves via the aorta and pulmonary artery. Semilunar valves then close so blood cannot return to the ventricles. Valves thus ensure one-way flow.

4. Phases of Heartbeat

- **Atrial systole:** both atria contract and push blood into the ventricles.
- **Ventricular systole:** ventricles contract, atrioventricular valves close, blood is pumped into the aorta and pulmonary artery.
- **General diastole:** the whole heart relaxes and fills with blood from the veins; the cycle repeats about 72 times per minute in a resting man.

5. LUB and DUB

LUB, the first, louder and longer sound, is produced when the atrioventricular (tricuspid and bicuspid) valves close at the start of ventricular contraction. DUB, the second, shorter and sharper sound, is produced when the semilunar valves close at the end of ventricular contraction. Heard with a stethoscope, these sounds indicate proper working of the heart valves.

6. Electrocardiogram (ECG)

The graphical record of the electrical activity of the heart during a cardiac cycle, measured by electrodes placed on the body; it records impulses originating from the SA node and spreading through the heart muscle.

- **P wave:** atrial depolarization and contraction.
- **QRS complex:** ventricular depolarization.
- **T wave:** ventricular repolarization and relaxation.

The shape and intervals of these waves show rate, rhythm and conduction of impulses from the SA node; any abnormality helps diagnose heart disorders.

Blood Vessels

LESSON 65

1. Types of Blood Vessels in Man

Human blood vessels are of three types: arteries, veins and capillaries. Each vessel type is structurally suited to its function in circulation.

Vessel	Structure	Function
Arteries	Thick, muscular and elastic walls with a narrow lumen	Carry blood away from the heart under high pressure
Veins	Thinner walls, a wider lumen and valves to prevent backflow	Carry blood towards the heart under low pressure
Capillaries	Wall of only one layer of endothelium, connecting arterioles and venules	Exchange of materials between blood and tissues

2. Capillaries and Exchange

Capillaries have walls of a single layer of endothelium so that exchange of materials can occur easily and rapidly. Oxygen, nutrients, hormones and waste products diffuse across this very thin wall between blood and tissue cells. A thicker wall would slow down this exchange. The huge number and total surface area of capillaries further increase the efficiency of exchange in the tissues.

3. Why Blood Flows Faster in Arteries

Blood flows faster in arteries because they receive blood directly from the pumping heart under high pressure. Arteries have narrower lumen relative to veins and thick elastic muscular walls that maintain this pressure. The total cross-sectional area of arteries is much less than that of capillaries, so velocity stays high. As blood reaches the wide capillary network and then veins, its speed and pressure decrease considerably.

Blood Pressure and Cardiovascular Disorders

LESSON 66

1. Blood Pressure in the Vessels

Arteries receive blood directly from the pumping heart under high pressure, and their thick elastic muscular walls maintain this pressure. Blood pressure falls continuously as blood travels away from the heart. As blood reaches the wide capillary network and then veins, its speed and pressure decrease considerably.

- **Lowest pressure:** Average blood pressure is lowest in the vena cava returning blood to the heart.
- **Pressure gradient:** Pressure is higher in the vessels nearer the heart, such as the aorta and arteries, and falls through arterioles and capillaries towards the vena cava.

2. Vasoconstriction

Vasoconstriction is the narrowing of the blood vessels. Stress and certain hormones (e.g. adrenaline) cause vasoconstriction, narrowing the blood vessels. Stress or hormone concentration is therefore a cause of vasoconstriction.

3. Blood Clotting at a Damaged Surface

Clotting is initiated when platelets and plasma factors come in contact with an irregular (damaged/injured) surface. Contact with an irregular surface by platelets and other factors in plasma is the event that initiates blood clotting.

4. Disorders and Detection

The shape and intervals of the ECG waves show the rate, rhythm and conduction of impulses from the SA node, and any abnormality in the ECG helps diagnose heart disorders. Damage to the SA node would disrupt the rate and timing of cardiac muscle contractions, showing why normal conduction is essential for a regular heartbeat. The heart sounds LUB and DUB indicate the proper working of the heart valves, so their study also reflects the working of the valves during ventricular contraction.

Lymphatic System of Man

LESSON 67

1. Lymph and Its Relation to Tissue Fluid

Lymph is essentially interstitial (tissue) fluid collected into lymph vessels, so it most closely resembles interstitial fluid. Lymph capillaries collect excess interstitial fluid from tissues and form lymph.

2. Parts of the Lymphatic System

The lymphatic system of man consists of lymph capillaries, lymph vessels, lymph nodes and lymphoid tissues such as the spleen.

- **Lymph capillaries:** Collect excess interstitial fluid and absorbed fats from tissues; in villi they are called lacteals.
- **Lymph vessels:** Lymph flows through larger vessels, which have one-way valves and are squeezed by nearby muscle contractions.
- **Lymph nodes:** Lymph passes through lymph nodes where it is filtered and lymphocytes are produced.
- **Lymphoid tissues:** Include tissues such as the spleen.

3. Return of Lymph to Blood

Finally lymph returns to the blood through the thoracic duct and the right lymphatic duct. The one-way valves in the lymph vessels and squeezing by nearby muscle contractions help lymph flow in one direction towards the blood.

4. Functions of the Lymphatic System

- **Fluid balance:** It maintains fluid balance by collecting excess interstitial fluid from tissues and returning it to the blood.
- **Fat absorption:** Lymph capillaries collect absorbed fats from tissues, and lacteals in the villi are involved in this collection.
- **Defence:** This system defends the body; lymph is filtered in lymph nodes and lymphocytes are produced there.

Together, the collection of interstitial fluid, its filtration through lymph nodes, and its return through the thoracic duct and right lymphatic duct keep tissue fluid balanced while supporting defence.

Immunity

UNIT 13

First Line of Defense

LESSON 68

1. Immunology and the Lines of Defense

The study of the functioning and disorders of the immune system is termed **immunology**. The body is defended in successive lines. The first line of defense prevents pathogens from entering the body and acts non-specifically against any pathogen trying to enter.

2. Physical Barriers

The **skin** and **mucous membrane** serve as physical barriers in the first line of defense. The digestive tract has both kinds of barriers: physical barriers in the form of its mucous lining, and biochemical barriers such as HCl, enzymes and saliva.

3. Biochemical Barriers

Chemical defences form part of the first line of defense. Six biochemical barriers of the body are:

- **Hydrochloric acid (HCl):** present in the stomach, it kills germs.
- **Lysozyme:** an enzyme in tears and saliva that dissolves bacterial walls.
- **Acidic secretions:** secretions of the skin and sweat check microbial growth.
- **Bile and digestive enzymes:** they destroy many pathogens entering with food.
- **Mucus:** it traps microbes.
- **Interferons and other antimicrobial proteins:** they inhibit viruses and bacteria.

Together, the physical and biochemical barriers stop most pathogens at the body surfaces before they can cause infection.

Second Line of Defense — Inflammation, Phagocytosis and Fever

LESSON 69

1. Ways of the Second Line of Defense

The second line of defense acts when pathogens cross the first line. These defences are non-specific but act rapidly:

- **Phagocytosis:** by neutrophils, monocytes/macrophages and dendritic cells.
- **Natural killer cells:** killing using perforin and granzymes.
- **Inflammation:** brings more blood and phagocytes to the infected area.
- **Fever:** inhibits microbial growth and speeds up defence reactions.
- **Antimicrobial proteins and complement:** they also attack invading microbes.

2. Phagocytosis

Phagocytosis is the process by which certain white blood cells engulf and digest pathogens and dead material. The phagocyte extends pseudopodia around the microbe, encloses it in a vacuole, and lysosomal enzymes kill and digest it. Both neutrophils and macrophages are phagocytes that engulf and destroy bacteria. The main WBCs acting as phagocytes are neutrophils, monocytes (which become macrophages in tissues) and dendritic cells. Eosinophils also show limited phagocytic action.

3. Inflammation

Inflammation is the local protective response of tissues to infection or injury, marked by redness, heat, swelling and pain. It develops because damaged cells, mast cells and basophils release histamine and other mediators that dilate blood vessels and increase their permeability, so plasma and phagocytes accumulate at the site. Phagocytes destroy the pathogens and clean the debris, helping healing; thus inflammation limits infection and starts tissue repair. It is controlled by anti-inflammatory mechanisms: once the pathogen is removed, mediator release stops, antihistamines and steroids can reduce excessive response, and macrophages clean the area. Controlled inflammation is protective, but excessive inflammation can damage tissues.

4. Fever — Mechanism and Protective Role

Fever develops when pyrogens released by pathogens and immune cells act on the temperature centre (hypothalamus) and raise its set-point. The body then produces and conserves more heat by shivering and vasoconstriction until the new temperature is reached. Protectively, raised temperature inhibits the growth of many bacteria and viruses and speeds up phagocytosis and immune reactions. Mild fever itself is protective.

5. Harmful Effects of Fever and Antipyretic Therapy

High and prolonged fever increases metabolic rate and water loss, causing dehydration, weakness and discomfort. Very high temperature can denature enzymes and damage tissues, especially of the brain. In children, sudden high fever may cause convulsions (fits). Prolonged fever also reduces appetite and can disturb heart and kidney function, so it must be controlled. **Antipyretic therapy** is the use of medicines that lower body temperature during fever, e.g. paracetamol. It is used when fever becomes very high or prolonged, because excessive fever can damage body proteins, cause discomfort, dehydration and convulsions, especially in children. By reducing the raised set-point of temperature in the hypothalamus, antipyretics bring relief, and are used mainly to control harmful effects of high fever.

Third Line of Defense — Antibodies and Cell-Mediated Immunity

LESSON 70

1. The Specific Immune Response

The third line of defense, consisting of T and B lymphocytes and antibodies, is responsible for the specific immune response. Any foreign substance that can elicit an immune response is called an **antigen**.

2. How a Bacterium Is Identified as Non-Self

A bacterium is identified as non-self because its surface carries antigens and molecular patterns different from the body's own markers. Antigen-presenting cells such as macrophages and dendritic cells engulf the bacterium and display its antigen fragments on their surface with MHC-II molecules. Helper T cells recognize this antigen-MHC combination as foreign, because body cells display only self antigens with MHC. This recognition activates the specific immune response. Self cells are protected because lymphocytes that react against self antigens are normally eliminated or inactivated. All nucleated cells exhibit **MHC-I** receptors, while MHC-II is found on specialized antigen-presenting cells like macrophages. Neutrophils are phagocytic but are not involved in antigen presentation.

3. Antibodies — Structure, Secretion and Differences

Antibodies are secreted by **plasma cells**, which are activated B cells transformed into antibody-producing cells. An antibody is a Y-shaped protein made of four polypeptide chains: two identical heavy chains and two identical light chains, joined by disulfide bonds. The stem and lower parts form constant regions, while the tips of the arms have variable regions that form two identical antigen-binding sites, specific for one antigen.

Even though the core proteins are the same, antibodies differ in their variable regions. The variable regions have different amino acid sequences, forming antigen-binding sites of different shapes, so each antibody binds a specific antigen. Antibodies also differ in their class (type of heavy chain), which determines their function and location, e.g. in blood, secretions or on mast cells.

4. How Antibodies Get Rid of Pathogens

Antibodies bind antigens, neutralizing toxins and viruses, clumping (agglutinating) microbes, and marking them for phagocytosis (opsonization). They also activate complement, which lyses the pathogen.

5. Natural Passive Immunity

Transfer of ready-made antibodies from mother to her fetus is **natural passive immunity**, as the fetus's own immune system does not produce them.

6. Cytotoxicity — NK Cell and Tc Cell

Cytotoxicity is the killing of target cells (infected or tumor cells) by immune cells through toxic substances. Both NK cells and Tc cells kill by releasing perforin, which makes pores in the target cell membrane, and granzymes, which enter through these pores and induce the target cell to die.

Feature	NK Cell	Tc Cell
Defense	Innate, non-specific, second line	Specific, third line, adaptive immunity
Activation	Without prior sensitization and without antigen presentation; acts immediately	Activated by specific antigens presented with MHC-I
Memory	Not antigen-specific, no memory	Antigen-specific, forms memory cells
Killing	Perforin and granzymes	Perforin and granzymes

The main difference is specificity and the need for activation: Tc cells are antigen-specific and have memory, NK cells are not.

7. How Tumour Cells Are Dealt With

The immune system constantly surveys the body for abnormal cells, including tumor cells, which display altered antigens on their surface. NK cells and cytotoxic T cells (Tc) recognize these abnormal cells as non-self and kill them by cytotoxicity, using perforin and granzymes. Macrophages also engulf dead tumor cells. This immune surveillance prevents many tumors from developing.

Autoimmune Disorders and Transplant Rejection

LESSON 71

1. Autoimmune Disorders

An autoimmune disorder is an abnormality in which the immune system fails to recognize one or more of its own tissues as self and attacks them, producing self-antibodies that destroy own cells. In all these disorders, the immune system fails to recognize its own tissues as self, and misdirected self-antibodies destroy the body's own cells.

2. Four Autoimmune Disorders of Man

- **Type-I diabetes mellitus:** immune cells destroy insulin-producing cells of the pancreas.
- **Rheumatoid arthritis:** joints are attacked causing pain and deformity.
- **Psoriasis:** a skin disorder with rapid abnormal skin growth.
- **Multiple sclerosis / myasthenia-type disorders:** nervous or muscle tissues are attacked.

Type-I diabetes and rheumatoid arthritis are also given as examples of autoimmune disorders in which self-antibodies destroy own cells.

3. Transplant Rejection — Role of T Cells and B Cells

In transplant rejection, the recipient's helper T cells recognize the donor organ as foreign through antigen-presenting cells, proliferate, and activate NK cells and cytotoxic T cells that directly attack the grafted tissue. This is cell-mediated rejection.

The activated helper T cells also signal B cells to become plasma cells whose antibodies destroy the graft. This is humoral rejection.

For this reason transplant patients are given immunosuppressant drugs.

Gaseous Exchange

UNIT 14

Respiration and the Human Respiratory System

LESSON 72

1. What Is Respiration?

Respiration is the process of exchanging gases with the environment and releasing energy from food; breathing is its mechanical part, consisting of inhalation and exhalation. The exchange of gases between the external environment and the lungs (blood at alveoli) is called external respiration.

2. How Breathing Developed in Animals

Breathing developed when animals became large and multicellular and simple diffusion through the body surface was no longer sufficient. Aquatic animals first developed gills for extracting oxygen from water. With the invasion of land, lungs and tracheal systems evolved to take oxygen directly from air, step by step with increasing body size, activity and terrestrial life.

3. Parts of the Human Respiratory System

The system comprises the nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles and lungs containing millions of alveoli, where oxygen diffuses into blood and CO2 diffuses out. The diaphragm, ribs and intercostal muscles bring about breathing movements, and the lungs are protected by pleural membranes. The trachea, supported by C-shaped cartilage rings, divides into two primary bronchi, one entering each lung, branching into smaller bronchi and bronchioles ending in alveoli. The right lung has three lobes, the left lung two lobes, and below the lungs lies the dome-shaped diaphragm.

4. Passage of Air to the Alveoli

External nares → nasal cavity → internal nares → pharynx → larynx → trachea → primary bronchi → secondary bronchi → tertiary bronchi → bronchioles → alveolar ducts → alveoli. Air enters through the mouth and nose and travels via the windpipe (trachea) to the lungs, filtered, warmed and moistened in the nasal cavity and trachea on its way.

5. Key Structures

- **Alveoli:** The tiny air sacs in human lungs; gaseous exchange takes place here, their thin moist walls surrounded by blood capillaries.
- **Nasal chambers:** Function as an air conditioner, warming, moistening and filtering incoming air.
- **Larynx:** A cartilage organ connecting pharynx to trachea; it houses the vocal cords and its epiglottis guards the trachea.
- **Trachea:** Conducts air and filters it through mucus and cilia, which trap dust and germs.

6. Three Regions of the Pharynx

The pharynx is a common passage for air and food.

- **Nasopharynx:** Behind the nasal cavity; only passes air; openings of the auditory (Eustachian) tubes equalize air pressure.
- **Oropharynx:** Behind the oral cavity; passes both air and food; contains the tonsils that help in defence.
- **Laryngopharynx:** Lowest part, leading to the larynx for air and the oesophagus for food; the epiglottis directs food away from the windpipe during swallowing.

7. Five Main Functions

- **Gaseous exchange:** Taking in oxygen and removing carbon dioxide.
- **Oxygen supply:** Supplying oxygen for cellular respiration and energy production.
- **Blood pH regulation:** Helping regulate blood pH by controlling CO2 level.
- **Sound production:** Through the vocal cords in the larynx.

- **Protection and smell:** Filtering, warming and moistening air, trapping germs in mucus, and sense of smell in the nasal cavity.

These functions keep the internal environment stable.

Mechanism of Breathing

LESSON 73

1. Type of Ventilation in Humans

Humans show pulmonary ventilation by negative pressure breathing (also described as tidal ventilation). Air is drawn into the lungs by expanding the chest cavity with the diaphragm and intercostal muscles, and the same pathway is used for air leaving the lungs. The same air passage therefore works in both directions, and fresh air mixes with some residual air in the lungs. Ventilation rate in a healthy adult is about 15–18 breaths per minute at rest.

2. Why It Is Called Negative Pressure Breathing

Human breathing is called negative pressure breathing because air is sucked into the lungs by creating a pressure inside the lungs lower than atmospheric pressure. When the diaphragm contracts and flattens and the rib cage expands, the volume of the chest cavity increases and pressure in the lungs falls below atmospheric pressure. Air therefore rushes in from outside to equalize the pressure. During expiration the process reverses and air is pushed out.

3. Inspiration and Expiration

Feature	Inspiration	Expiration
Type of process	Active process	Normally passive
Diaphragm and muscles	Diaphragm contracts and flattens; external intercostal muscles lift the ribs up and out	Diaphragm and intercostal muscles relax
Chest cavity volume	Increases	Decreases
Pressure in lungs	Falls below atmospheric pressure	Increased pressure in the lungs
Movement of air	Air rushes in through the air passage to the alveoli	Air is pushed out of the lungs

One complete inspiration and expiration forms one breath cycle.

4. Control of Breathing

Involuntary breathing is controlled by neurons located in the respiratory centres of the medulla and pons.

5. Why the Rate of Breathing Increases

The rate of breathing increases when the body needs more oxygen and produces more carbon dioxide, e.g. during exercise, exertion, fever or stress. Increased CO_2 and H^+ concentration in the blood is detected by chemoreceptors in the aortic and carotid bodies. They send signals to the respiratory centre in the medulla, which increases the rate and depth of breathing. This removes extra CO_2 and restores normal blood pH and oxygen supply.

6. Total Lung Capacity

The maximum volume of air contained in the lungs after a full forced inhalation is the total lung capacity.

Exchange and Transport of Gases

LESSON 74

1. Exchange of Gases at the Alveolar Level

At the alveolar level, exchange occurs by simple diffusion across the very thin (about 1 micrometre) barrier between alveolar air and blood capillaries. Oxygen concentration is higher in alveoli than in the incoming blood, so oxygen diffuses into the blood and binds with haemoglobin. Carbon dioxide concentration is higher in blood than in alveoli, so it diffuses out into the alveoli and is exhaled. The large surface area, moist surface and rich capillary network make this exchange very rapid and efficient. Gaseous exchange takes place in the alveoli, whose thin moist walls are surrounded by blood capillaries.

2. Haemoglobin — The Respiratory Pigment

Haemoglobin is called the respiratory pigment in humans and many animals. It is a red, iron-containing protein present in red blood cells. Each haemoglobin molecule can carry up to four oxygen molecules, forming oxyhaemoglobin, and transports oxygen from the lungs to the tissues. About 97% of oxygen in the body is transported by haemoglobin, while only about 3% is carried dissolved in plasma.

3. How CO2 Is Transported from Cells to the Lungs

Carbon dioxide produced in cells diffuses into blood capillaries and is transported to the lungs. Blood carries it through veins to the heart and then through the pulmonary artery to the lungs. At the alveoli, CO2 is released and exhaled.

Way of CO2 transport	Share	How it is carried
Dissolved in plasma	About 5-7%	A small part dissolves in the blood plasma
As carbaminohaemoglobin	About 20-23%	Binds with haemoglobin in red blood cells to form carbaminohaemoglobin
As bicarbonate ions	Major part, about 70%	Converted in RBCs, with the enzyme carbonic anhydrase, into carbonic acid which dissociates into bicarbonate ions (HCO_3^-) and H^+; bicarbonate is carried in plasma as $KHCO_3$ and $NaHCO_3$

At the lungs these reactions are reversed and CO2 is released for exhalation.

Respiratory Disorders

LESSON 75

1. Protection of the Air Passages

This chapter does not describe named respiratory diseases in detail. Its disorder-related content is about how the air passages are protected from dust, germs and infection, and how the entry of food and pathogens into the windpipe is prevented. When these defences work, the respiratory surface stays clean and the lungs are protected from infection and damage.

2. Hair, Mucus and Cilia

- **Hair in the nostrils:** Traps large dust particles and prevents them from entering the air passages.
- **Mucus glands:** Mucus glands in the nostrils and trachea secrete sticky mucus that traps fine dust, pollen and germs carried by inhaled air.
- **Cilia:** Cilia on the lining then sweep the mucus with trapped particles towards the pharynx, from where it is swallowed or expelled.
- **Trachea:** The trachea filters the air we breathe through its mucus and cilia, which trap dust and germs.
- **Nasal chambers:** The nasal chambers warm, moisten and filter the incoming air, functioning as an air conditioner.

Together, hair and mucus keep the respiratory surface clean and protect the lungs from infection and damage. Protection of air passages by filtering, warming and moistening air and trapping germs in mucus is one of the main functions of the respiratory system.

3. Guards Against Entry into the Windpipe

- **Epiglottis:** The epiglottis of the larynx guards the trachea, and in the laryngopharynx it directs food away from the windpipe during swallowing, leading air to the larynx and food to the oesophagus.
- **Tonsils:** The oropharynx contains the tonsils that help in defence.
- **Pleural membranes:** The lungs are protected by pleural membranes.

4. Breathing Change in Fever

The rate of breathing increases when the body needs more oxygen and produces more carbon dioxide, e.g. during exercise, exertion, fever or stress. Increased CO2 and H^+ concentration in the blood is detected by chemoreceptors, which signal the respiratory centre in the medulla to increase the rate and depth of breathing, removing extra CO2 and restoring normal blood pH and oxygen supply.